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Dimension- adaptive latent representation learning with normalized hyperbolic tensor rank for multi-view clustering

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ABSTRACT

In the field of multi-view clustering, latent representation methods have attracted much attention due to their ability to extract reliable feature representations from the underlying structure of raw data. However, existing latent representation methods often rely on pre-set parameter settings for selecting the dimension of the latent space, which lacks a robust theoretical foundation. Additionally, these methods typically fail to fully leverage the high-order correlations between views and overlook the structural features within individual views. To address these limitations, this paper proposes a Dimension-adaptive Latent Representation Learning with Normalized Hyperbolic Tensor Rank (DLRL-NHTR) for multi-view clustering. Firstly, a novel latent representation framework leveraging zero-space projection is introduced, effectively eliminating the reliance on manual dimension selection in previous methods. Furthermore, we propose an improved Normalized Hyperbolic Tensor Rank (NHTR) method to better capture high-order correlations between different views. Meanwhile, we also introduce an Elastic Structural Regularization (ESR) term that effectively captures multi-level structural information within individual views, thus enhancing the overall representation capabilities. Extensive experiments were conducted on ten benchmark datasets of varying types and scales, and the comparative results with 12 state-of-the-art methods demonstrate the superiority of our method in clustering performance.

1. Introduction

With the rapid advancement of information technology and the continuous expansion of data collection methods, an increasing number of real-world applications are producing data with multi-source and multi-modal characteristics, commonly referred to as multi-view data. For instance, in domains such as image processing, data mining, and social network analysis, features of different types or from different sources can describe the same object from multiple perspectives (Wang et al., 2026; Zhao et al., 2025b). Such multi-view data often contain complementary and richly structured information, offering stronger representational capacity and greater analytical value than single-view data. To fully exploit the shared information and the distinctive characteristics among views, multi-view clustering (MVC) has emerged as a research hotspot in machine learning and data mining (Gan et al., 2025; Yan et al., 2025). By performing joint representation learning on multi-view data and thoroughly exploring its underlying structure (Zhang et al., 2025; Zhao et al., 2025a), multi-view clustering not only improves clustering performance but also provides a more comprehensive understanding of the intrinsic patterns within the data.

The theoretical foundation of multi-view clustering is primarily based on two core principles: consistency and complementarity. The consistency principle holds the fact that although different views originate from diverse sources, they often share consistent intrinsic structural information when representing the same object or task. In contrast, the complementarity principle asserts that each view typically contains unique, discriminative information that other views cannot adequately capture. To effectively extract consistent structural information, researchers have developed a variety of clustering methods, utilizing techniques such as non-negative matrix factorization (NMF) (Cui et al., 2025), graph learning (Xu et al., 2024), spectral embedding (Ma et al., 2023), and subspace representation (Yu et al., 2023) to construct similarity graphs for each view. These graphs are then fused to form a unified consensus graph, which preserves the shared structural information while suppressing interference and noise from low-quality views. This fusion mechanism has been shown to substantially enhance the accuracy and robustness of multi-view clustering. However, relying solely on consistency information is insufficient to fully exploit the potential of multi-view data. Based on the complementarity principle, the unique information contained in each view is both crucial and indispensable for

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a comprehensive understanding of the data objects. To address this, researchers have proposed latent representation (LR) methods to uncover the complementary structures among views (Chen et al., 2020; Huang et al., 2022; Zhang et al., 2017). These methods typically assume that the intrinsic structure of multi-view data resides in a low-dimensional and shared latent subspace. By learning independent projection matrices for each view, LR methods map the original data into this common subspace, thereby obtaining a unified latent representation. LR methods not only strengthen the integration of consistent features but also facilitate effective information compensation and reconstruction across views, thereby fully harnessing the benefits of complementary features. For example, Zheng et al. (2023) used latent representations as the input to a degradation mapping model to mine complementary information across multiple views, thereby obtaining more accurate clustering results. Chen et al. (2021) further constructed a global similarity matrix based on the latent representation matrix to explore the global relationships among data samples. Wang et al. (2024) employed graph regularization to constrain low-dimensional latent representations, thereby better capturing the intrinsic similarity of multi-view data. Notably, Zhang et al. (2024) decomposed the latent representation into a common representation and a view-specific representation, and realized the coordinated modeling of consistency and complementarity in multi-view data by imposing orthogonality constraints. Although the above LR methods have advanced multi-view clustering in terms of consistency preservation, complementary information mining, and structural relationship representation, they still suffer from several limitations. First, the dimensionality of the latent representation usually needs to be manually predefined, making the model sensitive to parameter selection. Second, most existing methods mainly focus on modeling consistency and diversity, while remaining inadequate in addressing how to balance the relationship between global and local structures within each view.

In recent years, tensor representations have attracted substantial attention for their outstanding capability in capturing high-order correlations in multi-view data. The fundamental idea of these methods is to stack the representation matrices from each view into a third-order tensor and impose specific mathematical constraints to model high-order statistical dependencies among views. In this research field, Zhang et al. (2015) is the seminal work first introduced low-rank constraints into the tensor-based multi-view clustering (TMVC) framework, establishing the theoretical foundation for subsequent studies. Building on this work, later works such as leveraging tensor singular value decomposition (t-SVD) (Lu et al., 2016; Zhou et al., 2019) proposed representing the true tensor rank via the tensor average rank and rigorously proved that the tensor nuclear norm (TNN) (Pan et al., 2023b) serves as the tightest convex relaxation of this measure within a convex optimization framework. However, as the field of research developed, a limitation of TNN became apparent (Guo et al., 2022; Ji and Feng, 2025): its uniform treatment of all singular values can result in the over-penalization of large singular values and under-penalization of small ones, thereby compromising clustering performance. To address this limitation, several improved tensor rank approximation methods have emerged. For instance, Guo et al. (2022) introduced a weighted tensor Schatten-p norm that applies differentiated penalty parameters to singular values, yielding a more faithful approximation of the true rank. Zhong and Liu (2023) introduced a weighted non-convex low-rank norm to investigate the differences between various singular values. Ji and Feng (2025) proposed a Geman-function-based tensor rank surrogate to further improve the characterization of nonzero singular values. Pan et al. (2023a, 2024) developed different nonconvex tensor kernel norm to more accurately approximate the original tensor rank. Despite these efforts, accurately approximating tensor rank remains a nontrivial issue in tensor-based multi-view clustering. Existing studies have provided valuable insights into tensor rank characterization, while the design of more flexible and discriminative rank surrogates remains an open research topic. In particular, how to better reflect the contributions of different singular values to the underlying tensor structure is still worthy of further study.

Although existing latent representation and tensor-based multi-view clustering methods have achieved significant clustering performance, there are still several key issues that need to be addressed:

1. **Dimension selection problem.** The quality of latent representations is highly sensitive to the choice of latent space dimensions. An excessively large dimension may introduce noise, weakening the ability of the latent space to capture essential structures; conversely, an overly small dimension can cause information loss, hindering the extraction of complete semantic features. Most existing methods still rely on empirical settings, lacking theoretical guarantees.
2. **Tensor rank estimation problem.** Most existing TMVC methods often fail to impose sufficiently precise constraints on tensor singular values, thereby overlooking variations among singular values and leading to biased rank estimation. Thus, designing an improved tensor rank estimation technique that can effectively discriminate singular values and accurately approximate tensor rank remains an essential research direction.
3. **Structural representation problem.** Many TMVC methods emphasize high-order correlations across views while underexploring the multi-level structural information within individual views. In particular, they often overlook the balance between local nonlinear relationships and global linear structures. This fact may cause these methods to inadequately capture the complex distribution patterns of the original data, thereby degrading clustering performance.

Based on the above analysis, this paper proposes a novel tensor-based latent multi-view clustering framework, named Dimension-adaptive Latent Representation Learning with Normalized Hyperbolic Tensor Rank (DLRL-NHTR). First, unlike traditional latent representation learning methods that require a manually pre-defined latent dimensionality, we design a dimension-adaptive latent representation learning (DLRL) framework that automatically determines the latent subspace dimension according to the intrinsic rank information of the data. Specifically, DLRL exploits the null-space property to project the original data into a latent subspace determined by the intrinsic rank, rather than into a manually specified fixed-dimensional subspace adopted by most existing methods. This strategy fundamentally alleviates the performance fluctuations caused by dimension selection sensitivity in multi-view clustering. Second, we propose a new tensor rank approximation method, termed Normalized Hyperbolic Tensor Rank (NHTR), which employs a normalized hyperbolic function to impose adaptive and differentiated penalties on singular values of varying magnitudes, thereby constructing a more discriminative tensor rank constraint mechanism. Finally, to capture multi-level structural information within each view, an Elastic Structural Regularization (ESR) term is introduced, which enables our model to simultaneously perceive potential local non-linear features and global linear structures within individual views, thus achieving a more comprehensive and precise structural representation. To further highlight the novelty of the proposed method, the key differences between DLRL-NHTR and the most related multi-view clustering methods are summarized in Table 1.

The main contributions of this paper are summarized as follows:

1. A novel dimension-adaptive latent representation learning framework is proposed. Compared to traditional methods, DLRL avoids the influence of manually setting dimensional parameters and more effectively captures the latent structural features of the data.
2. A new tensor rank approximation method called Normalized Hyperbolic Tensor Rank is introduced. NHTR applies different penalty strengths to singular values of different magnitudes, allowing for precise differentiation between singular values of various scales. This improves the ability to analyze high-order correlations in multi-view clustering.
3. An elastic structural regularization is introduced to improve structural representation capabilities. ESR helps the proposed model capture both global and local structural features in views, yielding a

Table 1
Differences between our method and similar methods.

Methods	DIFF# 1	DIFF# 2	DIFF# 3
CMRL (Zheng et al., 2023)	✓/×	✓/×	✓/×
RMCLCS (Chen et al., 2021)	×/×	✓/×	✓/×
SSMC (Wang et al., 2024)	×/×	✓/×	✓/×
DEGREE (Zhang et al., 2024)	✓/×	✓/×	×/×
TPCH (Wang et al., 2025)	✓/✓	✓/×	✓/×
PFMSC (Ji and Feng, 2025)	✓/✓	×/×	×/×
Ours	✓/✓	✓/✓	✓/✓

*DIFF #1-#3 respectively represent whether the corresponding method is capable of capturing high-order correlations in data and distinguishing between different singular values; whether it is capable of learning latent representations and whether it avoids manually selecting the dimensions of latent representations; and whether it is capable of analyzing structural information in data and further modeling multi-level structural information.

more discriminative representation that faithfully characterizes the intrinsic structure of multi-view data.

4. An optimization algorithm based on the Augmented Lagrange multiplier method is proposed. Comparative experiments and ablation studies are conducted on ten benchmark datasets. The experiment results thoroughly validate the effectiveness and superiority of the proposed method.

The remaining sections of this paper are summarized as follows: In Section 2, we introduce the related methods. In Section 3, we introduced the proposed method in details, and then the optimization of our method and the complexity analysis are given in Section 4. In Section 5, we present our detailed experimental results and provide some analysis of our method. In the last section, we conclude this paper.

2. Related work

In the field of multi-view clustering, subspace-based MVC methods have attracted significant attention due to their strong theoretical foundation and mathematical interpretability. These methods are based on the assumption that the original data is composed of multiple linear subspaces (Tang et al., 2023; Wu et al., 2022). By learning the relationships between these subspaces, the intrinsic structural features of the data can be uncovered. The key to subspace methods is constructing a high-quality data representation matrix and applying specific regularization constraints to this matrix to achieve a clustering structure with good separability. Among these constraints, low-rank and sparsity constraints are the most commonly used regularization techniques, owing to their mathematical rigor and robustness to noise (Kong et al., 2025; Du et al., 2023). For instance, Luo et al. (2018) further decomposed the obtained representation matrix into two components: a consistency matrix and a diversity matrix, and imposed a low-rank constraint on the consistency matrix to ensure the shared information across multiple views. Lan et al. (2023) decomposed the representation matrix into three factors and utilized symmetry low-rank constraints to ensure a shared low-rank structure across views. It is worth mentioning that focusing on multi-level structural information in the view is beneficial for obtaining a better representation matrix. Lin et al. (2020) considered both global and local geometric information when constructing the representation matrix, thereby achieving an ideal feature representation. Hu et al. (2026) integrated structured sparsity and Laplacian regularization into a unified framework, establishing interdependence between the affinity matrix and cluster assignment. In addition, to reduce noise and outliers in the raw data, Zhang et al. (2017) firstly suggested that learning a latent representation from the raw data can address this issue. Meanwhile, latent representation learning can also thoroughly analyze the underlying structural information of data to obtain a comprehensive

structural representation. Upon the basis of this work, other related methods have been subsequently proposed (Zheng et al., 2023). Huang et al. (2022) combined graph learning with latent representations to adaptively learn low-dimensional representations that are resistant to noise interference and automatically generate similarity graphs with clear clustering structures. Shi et al. (2024) stacked representation matrices from different views to obtain augmented matrices and applied sparse constraints to enhance information. In addition, Zhao et al. (2024) integrated manifold representations from various views to provide consistency constraints for latent representations to improve clustering performance. Although latent representation methods have achieved remarkable progress in clustering, existing methods generally have a critical limitation: the dimension selection of the latent space still relies heavily on manual experience settings. This manual intervention not only introduces artificial errors but also restricts their adaptability and generalization performance. Recently, the TMVC method has gained great attention for its ability to capture high-order correlations across multiple views of data. For example, Lu et al. (2016) constructed a tensor structure by stacking and rotating representation matrices, and applied the TNN constraint to ensure low-rank consistency. Du et al. (2023) combined least squares methods with the TNN constraint to fully represent the consistency and complementarity of different representation matrices, effectively extracting underlying and higher-order information from the data. For existing non-linear relationships, Chen et al. (2019) combined kernel technology with high-dimensional mapping of raw data, capturing high-order relationships in the data while can analyze existing non-linear relationships. Existing studies have shown that TNN suffers from inherent limitations: it applies an equal treatment to all tensor singular values, which may lead to the excessive suppression of large singular values and inadequate constraints on small singular values. As noted in Ji and Feng (2025), different singular values contribute differently to data representation. Large singular values typically encode representative features and essential structural information, whereas small singular values are more likely to reflect noise or irrelevant details. For the TNN term, its uniform shrinkage strategy tends to over-contract large singular values, resulting in the loss of crucial structural information, while under-contracting small singular values, thereby allowing noise or redundant information to persist. Specifically, Xia et al. (2021) introduced a weighted singular value strategy that distinguishes the significance of different singular values by incorporating learnable parameters. Liu et al. (2019) innovatively applied the Schatten-p norm to multi-view clustering, enabling differentiated constraints on singular values. Ji and Feng (2025) proposed an enhanced tensor rank approximation method by systematically integrating the strengths of existing methods. While these methods achieve excellent performance in specific contexts, the development of more generalized tensor rank approximation techniques that can adapt to a wide range of practical applications remains a key challenge in current research.

3. Proposed method

The proposed method comprises two core components: dimension-adaptive latent representation learning (DLRL) and a normalized hyperbolic tensor rank (NHTR). DLRL, as the foundational framework of our method, overcomes the limitation of traditional methods that require a predefined latent space dimension, enabling the direct learning of latent representations from data. Once the latent representation is obtained, we obtain the representation matrices through self-representation learning and stack it into a tensor. Then, the NHTR and ESR constraints are applied. The novel tensor rank approximation NHTR strategy allows for more effective exploitation of high-order correlations within the data, and The ESR strategy can capture global and local structural information within individual view. The overall framework of the proposed method is illustrated in Fig. 1. Before introducing these two components in detail, we provide a systematic summary of the main mathematical sym-

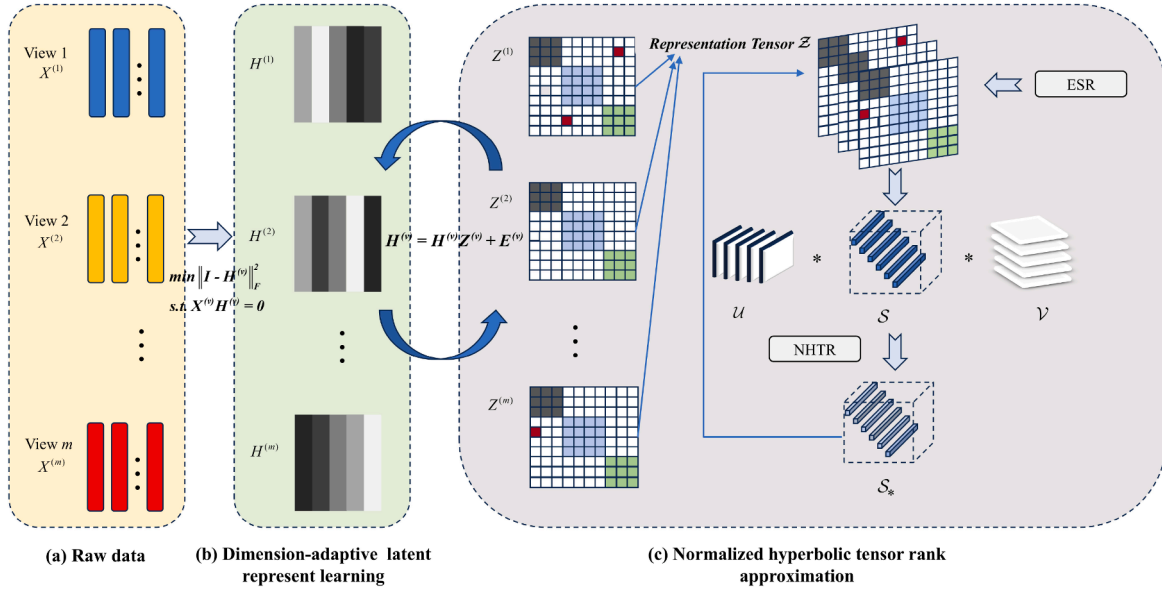


Fig. 1. The overall framework of the proposed method: (a) Raw data, from $X^{(1)}$ to $X^{(m)}$. (b) The dimension-adaptive latent representation learning to obtain $H^{(1)}$ to $H^{(m)}$. (c) Stacking the representation matrix $Z^{(v)}$ to obtain a tensor $\mathcal{Z}$, and apply Normalized Hyperbolic Tensor Rank (NHTR) and Elastic Structural Regularization (ESR) constraints.

Table 2

Notations and their descriptions.

Notation	Descriptions
$X^{(v)} \in \mathbb{R}^{d_v \times n}$	The v th view of multi-view data, where d_v and n denote the dimension and the number of the v th view, respectively.
$Z^{(v)} \in \mathbb{R}^{n \times n}$	Individual similarity graph of each view.
$E^{(v)} \in \mathbb{R}^{d_v \times n}$	Reconstruction errors of each view.
$Z_{:,i}^{(v)} (Z_{i,:}^{(v)})$	The i th column (row) vector of $Z^{(v)}$.
$\mathcal{Z} \in \mathbb{R}^{n \times n \times m}$	The third-order tensor.
$\mathbf{1}$	The vector in which all elements are 1.
$\mathbf{I}$	The identity matrix.

bolds and their definitions in Table 2 to ensure clarity and consistency in subsequent discussions.

3.1. Dimension-adaptive latent representation learning

Given a data $X^{(v)} \in \mathbb{R}^{d_v \times n}$, its null space is defined as $Null(X^{(v)}) = \{v^{(v)} \in \mathbb{R}^n \mid X^{(v)} v^{(v)} = 0\}$. In other words, the null space is the set of all column vectors $v^{(v)}$ that satisfy the matrix multiplication $X^{(v)} v^{(v)} = 0$. For $v^{(v)}$, we can use singular value decomposition (SVD) to obtain it, and the definition is as follows.

Definition 1 (The Null Space Vectors). Let $X^{(v)} \in \mathbb{R}^{d_v \times n}$, its singular value decomposition (Gai & Liu, 2025; Shi & Liu, 2023) is defined as

$$X^{(v)} = U^{(v)} \Sigma^{(v)} (V^{(v)})^T \quad (1)$$

where $U^{(v)} \in \mathbb{R}^{d_v \times d_v}$ and $V^{(v)} \in \mathbb{R}^{n \times n}$ are orthogonal matrices, $\Sigma^{(v)} \in \mathbb{R}^{d_v \times n}$ is a diagonal matrix. Let $\text{rank}(X^{(v)}) = r$. According to the Rank-Nullity theorem (Ji et al., 2014), the vectors corresponding to the $r^* = n - r$ singular values of $V^{(v)}$ form a standard orthogonal basis for $Null(X^{(v)})$. Thus, the null space vectors are defined as $N^{(v)} = \{v_1^{(v)}, v_2^{(v)}, \dots, v_{r^*}^{(v)}\} \in \mathbb{R}^{n \times r^*}$.

Based on the above definition, the latent representation is defined as

$$H^{(v)} = N^{(v)} (N^{(v)})^T \quad (2)$$

It is worth noting that although $H^{(v)}$ is an $n \times n$ matrix, it is essentially a projection matrix constructed from the null-space matrix $H^{(v)} = N^{(v)} (N^{(v)})^T$. The effective dimensionality of the latent representation is determined by $\text{rank}(H^{(v)}) = n - \text{rank}(X^{(v)}) = r^*$. Therefore,

there is no need to predefine the latent subspace dimension and the matrix $H^{(v)}$ automatically adapts its dimension based on the intrinsic rank of the data, achieving dimension-adaptive latent representation learning.

Moreover, $H^{(v)}$ exhibits the following two properties: First, as an orthogonal projection operator, it can effectively capture and encode the intrinsic structural information of the data. Second, under the subspace assumption, it yields an approximately block-diagonal structure, which further enhances the representation of subspace structures. In the following, we provide a detailed analysis of the two main properties of $H^{(v)}$.

Analysis of Property 1. Under the subspace assumption, the samples of $X^{(v)}$ generally lie on the union of several low-dimensional subspaces. Therefore, when the number of samples is sufficiently large, it usually holds that $\text{rank}(X^{(v)}) < n$. For samples belonging to the same subspace, they often satisfy identical or approximately similar linear constraint relationships, that is, there exists a nonzero vector $v^{(v)}$ such that $X^{(v)} v^{(v)} = 0$. This indicates that the intrinsic consistency among samples within the same cluster is reflected not only in their numerical similarity in the original feature space, but also in the shared linear constraint patterns shared by them. Therefore, characterizing sample relationships from the perspective of the null space can more directly capture the essential information associated with the underlying subspace structure. By definition of $N^{(v)}$, its column vectors form an orthonormal basis of $Null(X^{(v)})$, so $H^{(v)} = N^{(v)} (N^{(v)})^T$ satisfies $(H^{(v)})^T = H^{(v)}$, $(H^{(v)})^2 = H^{(v)}$, which means that $H^{(v)}$ is the orthogonal projection operator onto $Null(X^{(v)})$. Furthermore, for any $y \in \mathbb{R}^n$, we have

$$H^{(v)} y = \arg \min_{c \in Null(X^{(v)})} \|y - c\|_F^2 \quad (3)$$

which means $H^{(v)}y$ is the least-squares optimal approximation of any y in $\text{Null}(X^{(v)})$ (Appendix A provided its proof). This property indicates that $H^{(v)}y$ does not perform arbitrary mappings on the raw data, but rather preserves the effective components consistent with the structure within the structural subspace spanned by the null space in a manner that minimizes error. Simultaneously, it suppresses noise and interference terms deviating from this structural space, thereby reducing noise interference in the raw data.

Moreover, since $H^{(v)}$ is essentially a Gram-type matrix constructed from the nulls-space basis, its elements characterize pairwise inner product relationships between samples within this structural space. Consequently, it implicitly represents both structural consistency and separability among samples. For samples within the same subspace, sharing identical or nearly identical constraint patterns, their representations in the null space tend to be more similar, resulting in larger corresponding elements. Conversely, samples from different subspaces typically have smaller corresponding entries because their underlying constraint relationships are different. Leveraging this property, employing $H^{(v)}$ as a dictionary for self-representation learning highlights effective information related to clustering at the structural level. This approach mitigates interference from noise and outliers in representation learning, providing a beneficial foundation for subsequent tasks.

Analysis of Property 2. For property 2, we provide the following theorem for demonstration.

Theorem 1. The existing data $X^{(v)}$ is composed of c subspace sets (Kang et al., 2021, 2024), i.e. $X^{(v)} = \{X_1^{(v)}, X_2^{(v)}, \dots, X_c^{(v)}\}$, its latent representation $H^{(v)}$ has a block-diagonal structure.

Proof. According to (1), we redefine two linear products

$$\bar{P}^{(v)} = U^{(v)}(\Sigma^{(v)})^{\frac{1}{2}}, \quad \bar{Q}^{(v)} = (\Sigma^{(v)})^{\frac{1}{2}}(V^{(v)})^T \quad (4)$$

As reported in Costeira and Kanade (1998), we introduce an orthogonal matrix L and define $P^{(v)} = \bar{P}^{(v)}L$, $Q^{(v)} = L^{-1}\bar{Q}^{(v)}$, so that (1) can be rewritten as

$$X^{(v)} = P^{(v)}Q^{(v)} = (\bar{P}^{(v)}L)(L^{-1}\bar{Q}^{(v)}) \quad (5)$$

Combining the definition of the subspace assumption, we can obtain

$$\begin{aligned} X^{(v)} &= \{X_1^{(v)}, \dots, X_c^{(v)}\} \\ &= [P_1^{(v)}, \dots, P_c^{(v)}] \begin{bmatrix} Q_1^{(v)} & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & Q_c^{(v)} \end{bmatrix} \\ &= [\bar{P}_1^{(v)}, \dots, \bar{P}_c^{(v)}] \begin{bmatrix} L_1 & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & L_c \end{bmatrix} \begin{bmatrix} L_1^{-1} & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & L_c^{-1} \end{bmatrix} \begin{bmatrix} \bar{Q}_1^{(v)} & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & \bar{Q}_c^{(v)} \end{bmatrix}. \end{aligned} \quad (6)$$

According to (4)–(6), the following equation can be given

$$Q_*^{(v)} = L_*^{-1}\bar{Q}_*^{(v)} = L_*^{-1}\Sigma_*^{(v)\frac{1}{2}}(V_*^{(v)})^T \quad (7)$$

where

$$Q_*^{(v)} = \begin{bmatrix} Q_1^{(v)} & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & Q_c^{(v)} \end{bmatrix}, \begin{bmatrix} L_1^{-1} & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & L_c^{-1} \end{bmatrix}, \Sigma_*^{(v)} = \begin{bmatrix} \Sigma_1^{(v)} & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & \Sigma_c^{(v)} \end{bmatrix}, (V_*^{(v)})^T = \begin{bmatrix} (V_1^{(v)})^T & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & (V_c^{(v)})^T \end{bmatrix}.$$

Since $N^{(v)}$ is composed of some elements from the right singular value matrix, the following equivalent relationship exists.

$$H^{(v)} = N^{(v)}(N^{(v)})^T = V_*^{(v)}(V_*^{(v)})^T \quad (8)$$

From (7), we can observe that $V_*^{(v)} = (Q_*^{(v)})^T L_*^T (\Sigma_*^{(v)})^{-\frac{1}{2}}$, $(V_*^{(v)})^T = (\Sigma_*^{(v)})^{-\frac{1}{2}} L_* Q_*^{(v)}$, and $(\Sigma_*^{(v)})^{\frac{1}{2}} = L_* Q_*^{(v)} (V_*^{(v)})^{-T}$. So, (8) can be rewritten as

$$H^{(v)} = N^{(v)}(N^{(v)})^T = V_*^{(v)}(V_*^{(v)})^T = (Q_*^{(v)})^T L_*^T (\Sigma_*^{(v)})^{-1} L_* Q_*^{(v)}$$

$$\begin{aligned} &= (Q_*^{(v)})^T [L_*^{-1} (\Sigma_*^{(v)})^{\frac{1}{2}} (\Sigma_*^{(v)})^{\frac{1}{2}} L_*^{-T}]^{-1} Q_*^{(v)} \\ &= (Q_*^{(v)})^T [L_*^{-1} L_* Q_*^{(v)} (V_*^{(v)})^{-T} (V_*^{(v)})^{-1} (Q_*^{(v)})^T L_*^T L_*^{-T}]^{-1} Q_*^{(v)} \\ &= (Q_*^{(v)})^T [Q_*^{(v)} (Q_*^{(v)})^T]^{-1} Q_*^{(v)} \\ \text{let } Q_*^{(v)}(Q_*^{(v)})^T &= \begin{bmatrix} Q_1^{(v)} & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & Q_c^{(v)} \end{bmatrix} \begin{bmatrix} (Q_1^{(v)})^T & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & (Q_c^{(v)})^T \end{bmatrix} \\ &= \begin{bmatrix} \Delta_1^{(v)} & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & \Delta_c^{(v)} \end{bmatrix} \\ &= \begin{bmatrix} (Q_1^{(v)})^T & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & (Q_c^{(v)})^T \end{bmatrix} \begin{bmatrix} (\Delta_1^{(v)})^{-1} & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & (\Delta_c^{(v)})^{-1} \end{bmatrix} \\ &\quad \times \begin{bmatrix} Q_1^{(v)} & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & Q_c^{(v)} \end{bmatrix} \\ &= \begin{bmatrix} \Theta_1^{(v)} & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & \Theta_c^{(v)} \end{bmatrix} \end{aligned} \quad (9)$$

Obviously, $H^{(v)}$ is a block-diagonal matrix. $\square$

According to Theorem 1, samples belonging to the same subspace satisfy identical or approximately similar constraint patterns in the null space. As a result, samples tend to exhibit more consistent structural relationships after null-space projection, and the resulting projection matrix is more likely to present a block-diagonal structure. From the perspective of subspace clustering, this property is particularly important, because the key to revealing clustering structure lies not in characterizing the directions of strongest overall variation in the data, but in identifying which samples share the same underlying subspace constraints.

In contrast, representations based on the row space or column space generally emphasize the dominant variation directions of the data. Such directions are often the joint outcome of multiple subspaces and can be easily influenced by large clusters or noise, thereby blurring the boundaries between different subspaces. Consequently, these representations are more suitable for describing the global geometric shape of the data, but may not directly reveal a clear cluster structure. Furthermore, compared with other common decomposition methods, null-space projection is also more consistent with the objective of subspace clustering. For example, NMF mainly focuses on part-based representations through nonnegative weighted combinations, which is advantageous for data reconstruction and interpretability, but is relatively less sensitive to subspace structures characterized by shared constraint relationships; Laplacian feature spaces rely on a preconstructed similarity graph, whose quality is susceptible to noise and variations in view quality. Overall, under the subspace assumption, null-space projection more directly reflects the shared linear constraint relationships among samples and is more capable of preserving the separability among subspaces, making it more suitable as a structured dictionary for self-representation learning.

Thus, using $H^{(v)}$ as the dictionary for self-representation learning offers several advantages. First, $H^{(v)}$ mitigates the interference of noise and outliers in self-representation learning. Second, since the self-representation model is defined as $H^{(v)} = H^{(v)}Z^{(v)} + E^{(v)}$, $H^{(v)}$ and $Z^{(v)}$ exist within the same structural space. Moreover, the effective dimensionality of this space is adaptively determined by $\text{rank}(H^{(v)}) = n - \text{rank}(X^{(v)})$, which enables the representation space to better match the intrinsic subspace structure of the data. This not only reduces redundant degrees of freedom and cross-cluster spurious connections, but also strengthens the property of “high representability within clusters and low representability across clusters”. In addition, under the subspace assumption, the theoretical analysis shows that $H^{(v)}$ exhibits a block-diagonal structure. This indicates that when employed as a dictionary, $H^{(v)}$ naturally preserves the separability among subspaces, thus provid-

ing a favorable foundation for learning a self-representation coefficient matrix with a clear structural pattern.

To facilitate the solution of (2), we give the following Theorem.

Theorem 2. For $H^{(v)}$, its general solution can be obtained from the following optimization problem:

$$\min \|I - H^{(v)}\|_F^2 \text{ s.t. } X^{(v)}H^{(v)} = 0 \quad (10)$$

Let $H^{(v)}$ be the matrix whose columns form a standard orthonormal basis of $\text{Null}(X^{(v)})$, i.e., $X^{(v)}N^{(v)} = 0$ and $(N^{(v)})^T N^{(v)} = I$. Then the optimal solution to the above optimization problem is

$$H^{(v)} = N^{(v)}(N^{(v)})^T \quad (11)$$

Proof. From the constraint $X^{(v)}H^{(v)} = 0$, it follows that each column of $H^{(v)}$ lie in $\text{Null}(X^{(v)})$. Since the columns of $H^{(v)}$ form a basis of $\text{Null}(X^{(v)})$, any feasible solution $H^{(v)}$ can be expressed as $H^{(v)} = N^{(v)}\Psi$, where Ψ is a free variable. Substituting this into the constraint gives $X^{(v)}H^{(v)} = X^{(v)}N^{(v)}\Psi = 0$, which holds for all feasible. Hence, the original problem is equivalent to the unconstrained problem:

$$\min \|I - N^{(v)}\Psi\|_F^2 \quad (12)$$

Expanding the objective function in (12) yields

$$\begin{aligned} \|I - N^{(v)}\Psi\|_F^2 &= \text{tr}((I - N^{(v)}\Psi)^T(I - N^{(v)}\Psi)) \\ &= \text{const} - 2\text{tr}(N^{(v)}\Psi) + \text{tr}(\Psi^T N^{(v)T} N^{(v)}\Psi) \end{aligned} \quad (13)$$

where const denotes a constant term. Since $N^{(v)T} N^{(v)} = I$, (13) simplifies to

$$\|I - N^{(v)}\Psi\|_F^2 = \text{const} - 2\text{tr}(N^{(v)}\Psi) + \text{tr}(\Psi^T \Psi) \quad (14)$$

Taking the derivative with respect to Ψ and setting it to zero, we can obtain $\Psi = (N^{(v)})^T$. Thus, Theorem 2 is proved.

To obtain more comprehensive information about the data, we employ the idea of self-representative subspace learning to obtain the representation matrix $Z^{(v)}$ for each view, as follows

$$\begin{aligned} \min \sum_{v=1}^m \|I - H^{(v)}\|_F^2 + \alpha \|E\|_{2,1} \\ \text{s.t. } X^{(v)}H^{(v)} = 0, v \in 1, 2, \dots, m, \\ H^{(v)} = H^{(v)}Z^{(v)} + E^{(v)}, Z^{(v)} \geq 0, \\ E = [E^{(1)}; E^{(2)}; \dots; E^{(m)}] \end{aligned} \quad (15)$$

where $E^{(v)}$ means the reconstruction error. From (15), it avoids the problem of dimension selection in previous latent representation learning and provides theoretical support. $\square$

3.2. Normalized hyperbolic tensor rank

The mainstream idea in tensor-based multi-view methods is to stack multiple representation matrices obtained through self-representation learning into a third-order tensor and then apply the TNN constraint to capture high-order correlations across different views. Unfortunately, TNN constrains singular values through simple summation. Small singular values of a tensor generally correspond to irrelevant or noisy information, whereas large singular values typically reflect essential structural information. However, TNN fails to differentiate its relative importance. Consequently, inspired by Gu and Feng (2024), we give an effective tensor rank approximation method, called Normalized Hyperbolic Tensor Rank (NHTR). It applies stronger penalties to small singular values to suppress their negative effects, while imposing weaker penalties on large singular values to preserve and exploit their valuable information. Thus, our proposed NHTR is defined in the following form.

Definition 2. First, we define the normalized hyperbolic tangent function.

$$\text{Nortanh}(x) = \frac{\tanh(\gamma x)}{\tanh(\gamma)} = \frac{(e^{\gamma x} - e^{-\gamma x})(e^{\gamma} + e^{-\gamma})}{(e^{\gamma x} + e^{-\gamma x})(e^{\gamma} - e^{-\gamma})} \quad (16)$$

So, for a given tensor $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, its Normalized Hyperbolic Tensor Rank is represented as

$$\begin{aligned} \|\mathcal{A}\|_{\text{NHTR}} &= \frac{1}{n_3} \sum_{k=1}^{n_3} \|\mathcal{A}_f^k\|_{\text{NHTR}} = \text{Nortanh}(S_f^k(i, i)) \\ &= \frac{1}{n_3} \sum_{k=1}^{n_3} \sum_{i=1}^h \left(\frac{(e^{\gamma S_f^k(i, i)} - e^{-\gamma S_f^k(i, i)})(e^{\gamma} + e^{-\gamma})}{(e^{\gamma S_f^k(i, i)} + e^{-\gamma S_f^k(i, i)})(e^{\gamma} - e^{-\gamma})} \right) \end{aligned} \quad (17)$$

where $\gamma > 0$, $h = \min(n_1, n_2)$, and S is obtained by t-SVD of $\mathcal{A}_f = \mathcal{U}_f \mathcal{S}_f \mathcal{V}_f^T$ in the Fourier domain.

To evaluate the effectiveness of the proposed NHTR method and distinguish its differences from other related approaches, we conducted a comparative analysis of NHTR with two classical tensor rank approximation methods TNN and TLSpN and the recently proposed TGR method. The corresponding approximation functions of these methods are defined as $f_{\text{TNN}}(x) = x$, $f_{\text{TLSpN}}(x) = \log(1 + x^p)$, $f_{\text{TGR}}(x) = \frac{x}{\eta + x}$, $f_{\text{NHTR}}(x) = \left(\frac{e^{\gamma x} - e^{-\gamma x}}{e^{\gamma x} + e^{-\gamma x}} \right) \left(\frac{e^{\gamma} + e^{-\gamma}}{e^{\gamma} - e^{-\gamma}} \right)$, and the results are presented in Fig. 2, where the x-axis represents the singular values and the y-axis denotes the approximated values. The experimental results in Fig. 2 indicate that different rank approximation functions exhibit distinct behaviors across various ranges of singular values. To more comprehensively characterize the properties of each method, a detailed analysis can be provided in the following three aspects:

First, when the singular value is zero, the true rank should be zero. As shown in Fig. 2, TNN, TLSpN (Guo et al., 2022), TGR (Ji and Feng, 2025), and NHTR all satisfy this condition. This indicates that the aforementioned methods can correctly capture the fundamental property that zero singular values have no contribution to the rank.

Second, for large singular values, the true rank is expected to approach a numerical value of 1. In this regard, the methods exhibit notable differences. The approximation results of TNN increase linearly with the singular values, whereas TLSpN grows more slowly but still continues to increase as the singular values become larger, i.e., $x \rightarrow \infty$, $f_{\text{TNN}}(x)$, $f_{\text{TLSpN}}(x) \gg 1$. This implies that both methods continuously impose penalties in the region of large singular values, which can lead to excessive compression of the primary structural information and hinder accurate rank estimation. In contrast, TGR and NHTR are able to adaptively reduce the penalty on large singular values, achieving approximations that are closer to the true rank, i.e., $x \rightarrow \infty$, $f_{\text{TGR}}(x)$, $f_{\text{NHTR}}(x) = 1$. Specifically, the proposed NHTR method effectively mitigates the penalty on large singular values that represent the dominant structural components, a property it shares with the recently TGR method.

Furthermore, in the intermediate singular value region, the differences among these methods become even more noteworthy. Taking the reference point (0.10, 0.91) in Fig. 2 as an example for detailed analysis: On the left side of this point $0 < x < 0.9$, corresponding to the smaller singular value range, the adaptive penalty functions of TGR and NHTR yield approximations that are clearly closer to the true rank compared to TNN and TLSpN. This indicates that NHTR aligns with the recently TGR method by imposing stronger penalties on smaller singular values, thereby achieving more accurate rank approximation. Notably, within this range, TGR slightly outperforms NHTR in approximating the true rank, primarily due to differences in their local penalty intensities. Specifically, the first-order derivative of TGR is approximately $f'_{\text{TGR}}(x) \sim (x + \eta)^{-2}$, while that of NHTR is approximately $f'_{\text{NHTR}}(x) \sim e^{-2\gamma x}$. The first-order derivative reflects the local penalty strength, i.e., the rate at which the penalty function increases with an infinitesimal increase in the singular value at a given point. Because TGR's local penalty grows quadratically, it has a clear advantage when handling smaller singular values. As shown in the Fig. 2, by tuning the parameters of NHTR, its local approximation performance can approach or even match that of TGR. On the right side of the reference point $x \geq 0.9$, corresponding to larger singular values, the region is magnified in Fig. 2 for closer inspection.

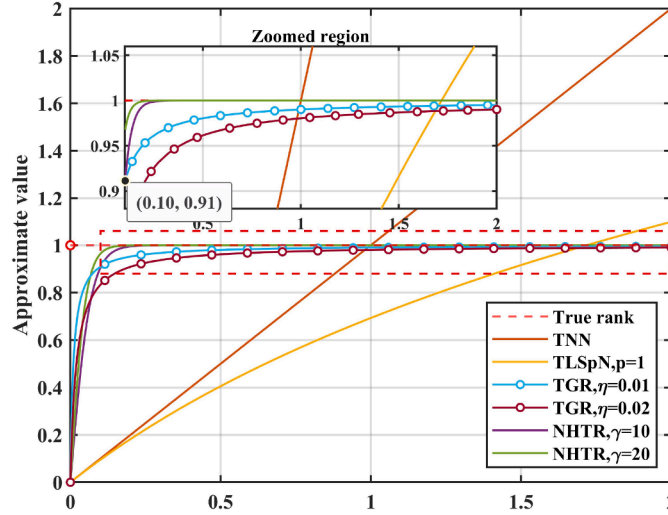


Fig. 2. Comparison of NHTR with different methods for approximating the true rank function.

tion. The magnified view shows that NHTR clearly outperforms TGR, owing to its exponentially decaying local penalty. As the singular values increase, the additional penalty imposed by NHTR diminishes more rapidly, allowing it to effectively avoid continuous compression of the useful singular values. This property enables NHTR to better preserve the main structural information in the data.

In summary, the main differences among these methods lie in their local penalty mechanisms. TNN and TLSpN are essentially monotonically increasing approximation functions and lack effective local penalties. While they can partially capture low-rank characteristics, they struggle to accurately reflect the behavior of the true rank. In contrast, both TGR and NHTR employ adaptive penalty mechanisms that account for differences in singular values. Specifically, TGR's penalty intensity decays quadratically, rising rapidly in the region of very small singular values, which provides an advantage in local approximation. However, it continues to impose relatively strong penalties on larger singular values, potentially leading to excessive compression of dominant structural components. In contrast, NHTR not only achieves effective approximation of large singular values but also features a local penalty that decays rapidly as singular values increase. This faster reduction in additional penalties better preserves the information of significant singular values. Consequently, NHTR demonstrates superior overall performance, balancing accurate approximation for small singular values with protection of large singular values, and outperforms TGR, TNN, and TLSPN in overall rank approximation.

Then, we combine (15) and (17) to obtain the following function

$$\begin{aligned} \min \quad & \|Z\|_{NHTR} + \sum_{v=1}^m \|I - H^{(v)}\|_F^2 + \alpha \|E\|_{2,1} \\ \text{s.t.} \quad & X^{(v)} H^{(v)} = 0, v \in 1, 2, \dots, m, \\ & H^{(v)} = H^{(v)} Z^{(v)} + E^{(v)}, Z^{(v)} \geq 0, \\ & Z = \Phi(Z^{(1)}, Z^{(2)}, \dots, Z^{(m)}), \\ & E = [E^{(1)}; E^{(2)}; \dots; E^{(m)}] \end{aligned} \quad (18)$$

In addition, to fully exploit the multi-structural information embedded in each view and to more comprehensively capture the intrinsic structural characteristics of the data, we innovatively introduce an elastic structural regularization (ESR) term, which is defined as:

$$\|Z\|_{ESR} = \sum_{v=1}^m \|Z^{(v)}\|_{ESR} = \sum_{v=1}^m \text{tr}((Z^{(v)})^T ((1-v)L^{(v)} + vM) Z^{(v)}) \quad (19)$$

where $L^{(v)} = D^{(v)} - W^{(v)}$ is the Laplacian matrix of $W^{(v)}$, which is used to reveal local manifold information in the v th view, and $W^{(v)}$ is obtained

by solving following problem

$$\min_{W_{:,i}^{(v)} \mathbf{1} = 1, W_{:,i}^{(v)} > 0} \sum_{i,j=1}^n \|X_{:,i}^{(v)} - X_{:,j}^{(v)}\|_F^2 W_{i,j}^{(v)} + \lambda \sum_{i=1}^n \|W_{:,i}^{(v)}\|_F^2 \quad (20)$$

the solution has been given as (Li et al., 2020). $D_{i,i}^{(v)} = \sum_{j=1}^n W_{i,j}^{(v)}$ is the

degree matrix. $M = \frac{1}{n} \mathbf{1}_n \mathbf{1}_n^T - I$ is a linear correlation matrix that represents the overall degree of linear correlation in the v th view, and v is the balance parameter. Next, we will explain in detail how to obtain this regularization.

As reported in Wei et al. (2023), a transitive relationship exists between the raw data and its corresponding representation matrix. In other words, if a certain relationship is observed between samples in the raw data, this relationship will also be reflected in the representation matrices of these samples. Consequently, the local correlation matrix is equally applicable to the representation matrix $Z^{(v)}$.

$$\begin{aligned} \frac{1}{2} \sum_{i,j} \|Z_{:,i}^{(v)} - Z_{:,j}^{(v)}\|_F^2 W_{i,j}^{(v)} &= \text{tr}((Z^{(v)})^T D^{(v)} Z^{(v)} \\ &\quad - (Z^{(v)})^T W^{(v)} Z^{(v)}) = \text{tr}((Z^{(v)})^T L^{(v)} Z^{(v)}) \end{aligned} \quad (21)$$

As a classic unsupervised method, Principal Component Analysis (PCA) (Bro & Smilde, 2014) maximizes the covariance matrix of the data to characterize its global linear structure. Inspired by PCA, we propose maximizing the structured covariance matrix of the multi-view representation matrices, thereby systematically capturing the global linear relationships embedded within the v th view, as follows

$$\begin{aligned} S &= \frac{1}{n} (Z^{(v)} - \bar{Z}^{(v)}) (Z^{(v)} - \bar{Z}^{(v)})^T \\ &= \frac{1}{n} (Z_{:,1}^{(v)} - \bar{Z}^{(v)}, \dots, Z_{:,n}^{(v)} - \bar{Z}^{(v)}) (Z_{:,1}^{(v)} - \bar{Z}^{(v)}, \dots, Z_{:,n}^{(v)} - \bar{Z}^{(v)})^T \end{aligned} \quad (22)$$

where $\bar{Z}^{(v)} = \frac{1}{n} \sum_{i=1}^n Z_{:,i}^{(v)}$. For $(Z_{:,1}^{(v)} - \bar{Z}^{(v)}, \dots, Z_{:,n}^{(v)} - \bar{Z}^{(v)})$, it can be rewritten as

$$\begin{aligned} &(Z_{:,1}^{(v)} - \bar{Z}^{(v)}, \dots, Z_{:,n}^{(v)} - \bar{Z}^{(v)}) \\ &= (Z_{:,1}^{(v)}, \dots, Z_{:,n}^{(v)}) - (\bar{Z}^{(v)}, \dots, \bar{Z}^{(v)}) = (Z^{(v)})^T - \bar{Z}^{(v)} (\mathbf{1}, \dots, \mathbf{1}) \\ &= (Z^{(v)})^T - \frac{1}{n} (Z^{(v)})^T \mathbf{1}_n \mathbf{1}_n^T = (Z^{(v)})^T (I - \frac{1}{n} \mathbf{1}_n \mathbf{1}_n^T) \end{aligned} \quad (23)$$

Thus, (23) is equal to

$$\begin{aligned} S &= \frac{1}{n} (Z^{(v)} - \bar{Z}^{(v)}) (Z^{(v)} - \bar{Z}^{(v)})^T \\ &= \frac{1}{n} (Z_{:,1}^{(v)} - \bar{Z}^{(v)}, \dots, Z_{:,n}^{(v)} - \bar{Z}^{(v)}) (Z_{:,1}^{(v)} - \bar{Z}^{(v)}, \dots, Z_{:,n}^{(v)} - \bar{Z}^{(v)})^T \\ &= \frac{1}{n} (Z^{(v)})^T (I - \frac{1}{n} \mathbf{1}_n \mathbf{1}_n^T) (I - \frac{1}{n} \mathbf{1}_n \mathbf{1}_n^T) Z^{(v)} \end{aligned} \quad (24)$$

Obviously, $I - \frac{1}{n} \mathbf{1}_n \mathbf{1}_n^T$ has symmetry and idempotency. By minimizing (21) and maximizing (24), and introducing the balance parameter

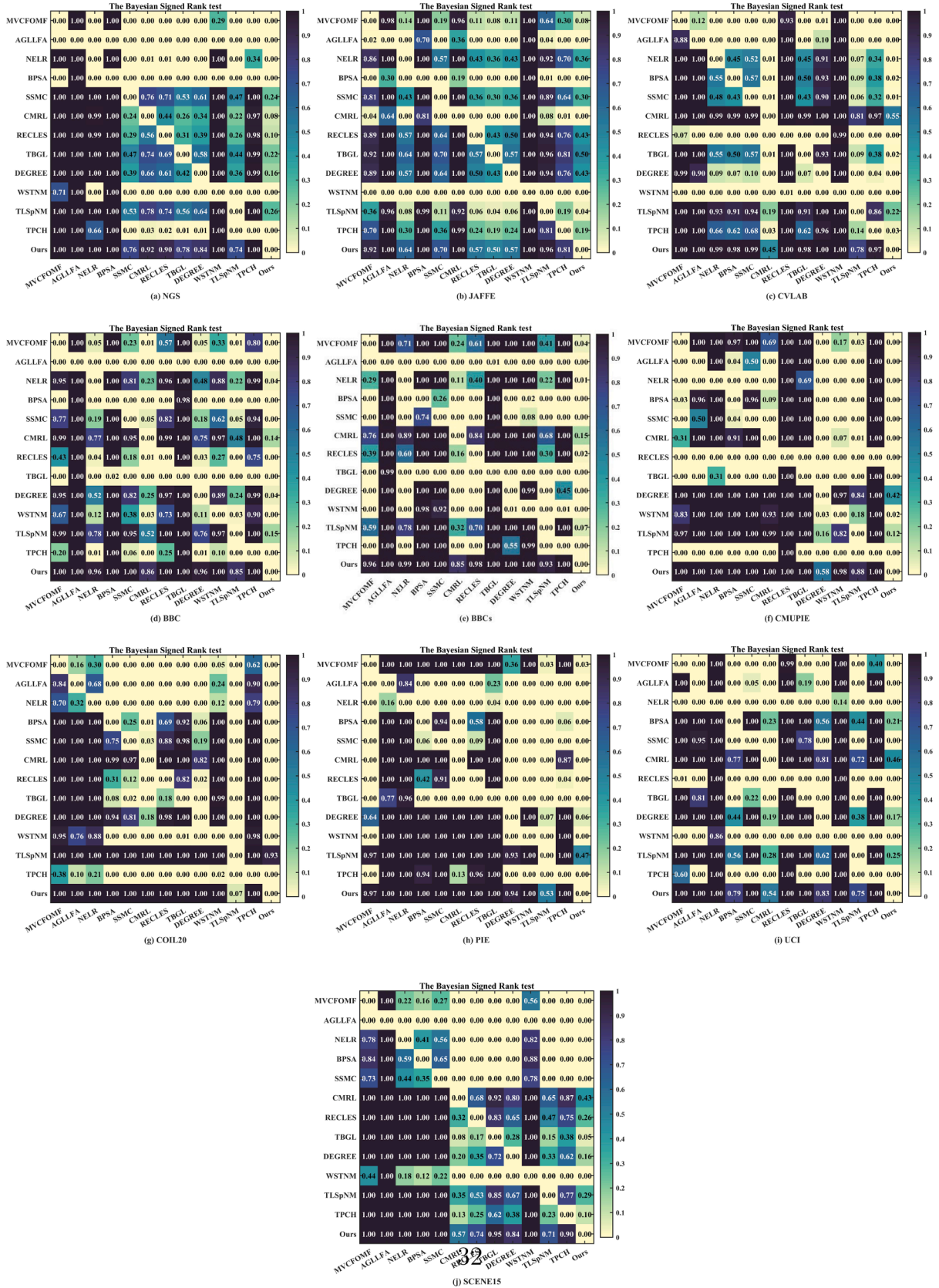
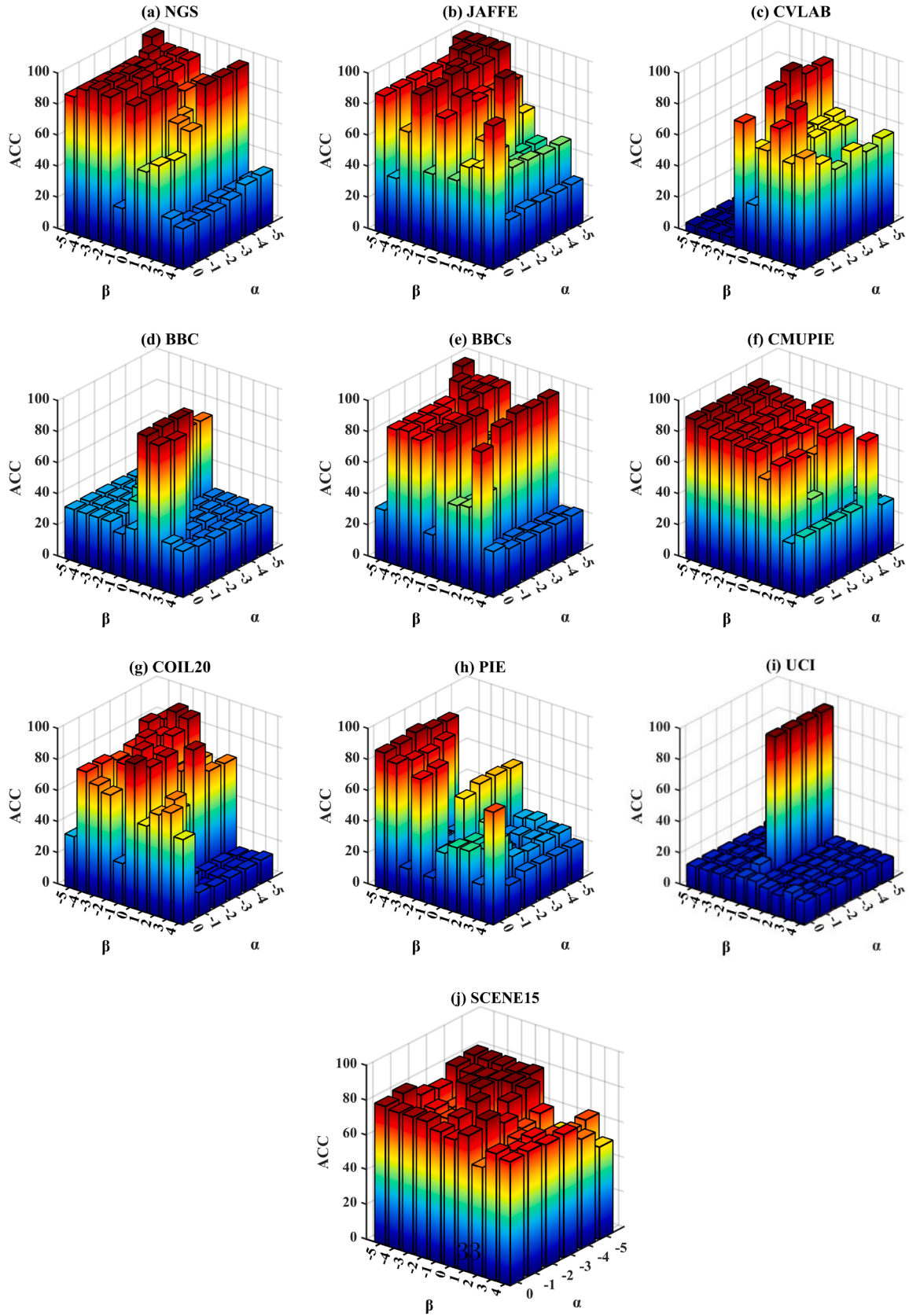


Fig. 3. The Bayesian signed rank test between different methods across all datasets.

Fig. 4. The parameter α and β vs ACC on different datasets.

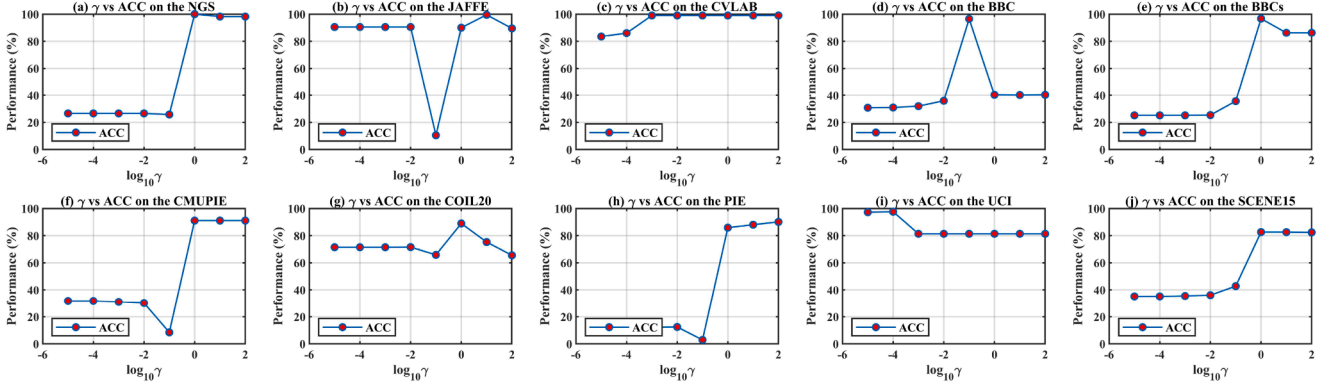
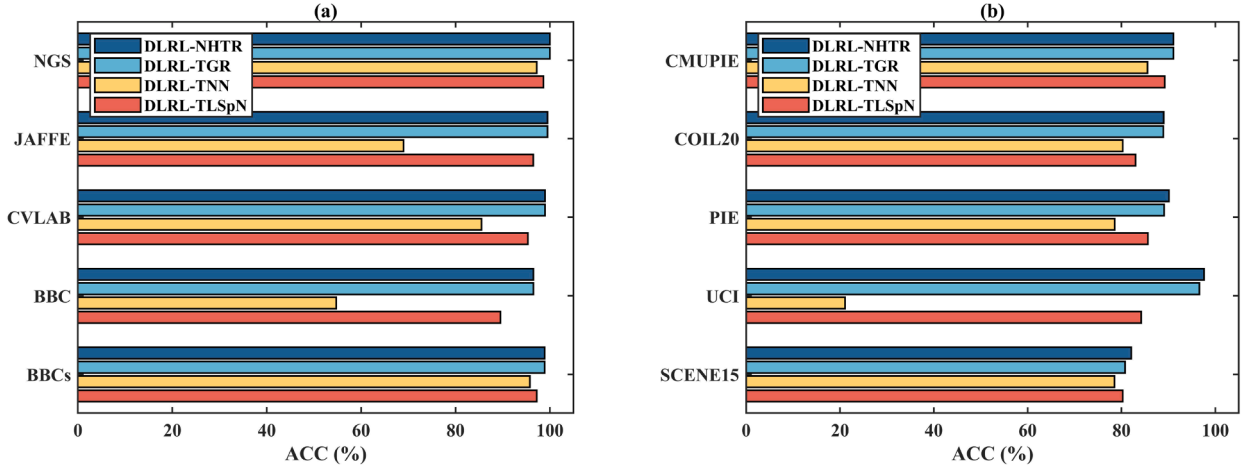
Fig. 5. The rank parameter γ vs ACC on different datasets.

Fig. 6. Clustering performance of different rank function on difference datasets: (a) NGS, JAFFE, CVLAB, BBC, and BBCs; (b) CMUPE, COIL20, PIE, UCI, and SCENE15.

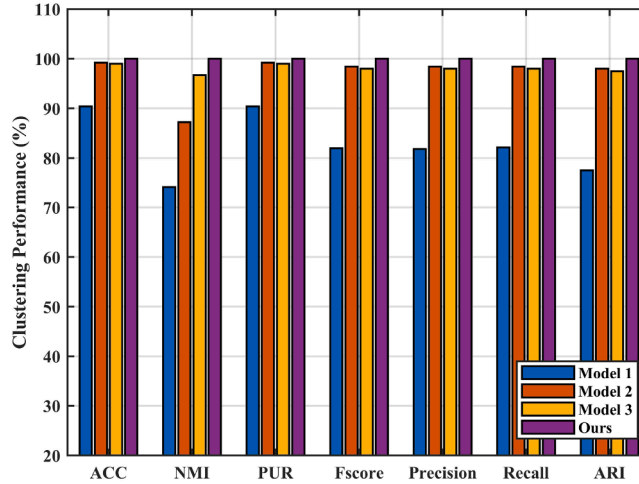


Fig. 7. Comparison of clustering metrics for the proposed method and its variants on NGS datasets.

v , the ESR can be obtained. Combining (18) and (19), the final function can be given as

$$\begin{aligned}
 \min \quad & \|Z\|_{NHTR} + \sum_{v=1}^m \left\| I - H^{(v)} \right\|_F^2 + \alpha \|E\|_{2,1} + \|Z\|_{ESR} \\
 s.t. \quad & X^{(v)} H^{(v)} = 0, v \in 1, 2, \dots, m, \\
 & H^{(v)} = H^{(v)} Z^{(v)} + E^{(v)}, Z^{(v)} \geq 0, \\
 & Z = \Phi(Z^{(1)}, Z^{(2)}, \dots, Z^{(m)}), \\
 & E = [E^{(1)}; E^{(2)}; \dots; E^{(m)}]
 \end{aligned} \tag{25}$$

The main advantages of the proposed model are as follows: (1) Unlike previous methods, the model does not require manual dimension selection and can directly obtain the latent representations of multiple views. (2) Theoretical analysis shows that the learned latent representation exhibits a block-diagonal structure. This property facilitates the formation of more realistic and separable cluster boundaries in the representation matrix $Z^{(v)}$, thereby enhancing clustering performance. (3) The model achieves a more accurate tensor rank approximation, which significantly improves its ability to analyze higher-order correlations

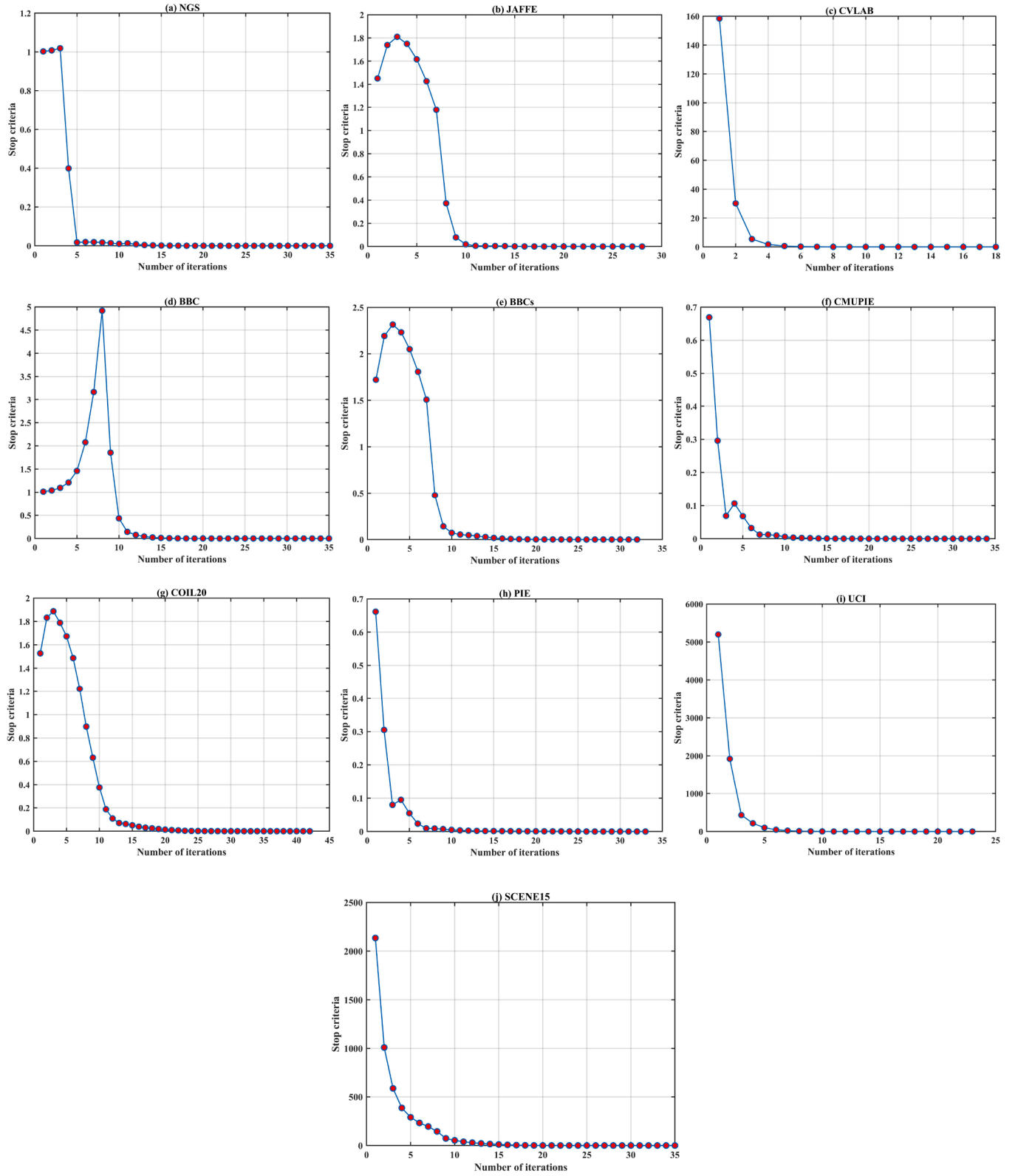


Fig. 8. The curve of convergence results on different datasets.

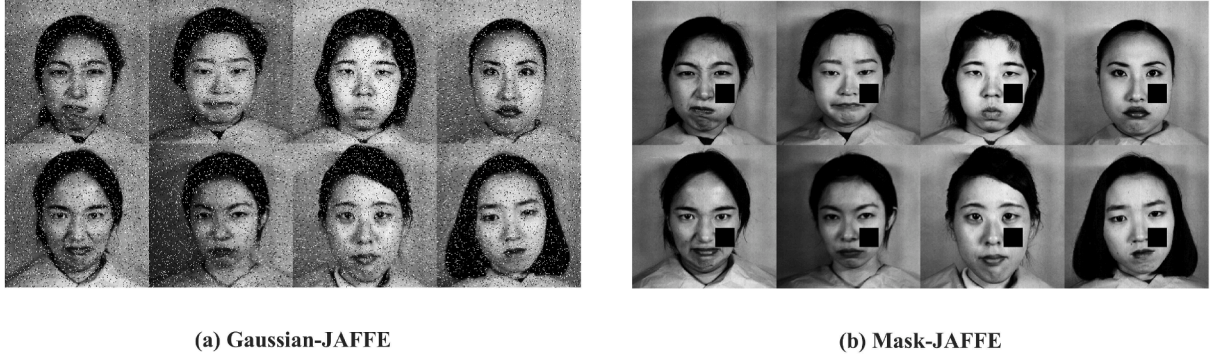


Fig. 9. Different examples of two noisy datasets.

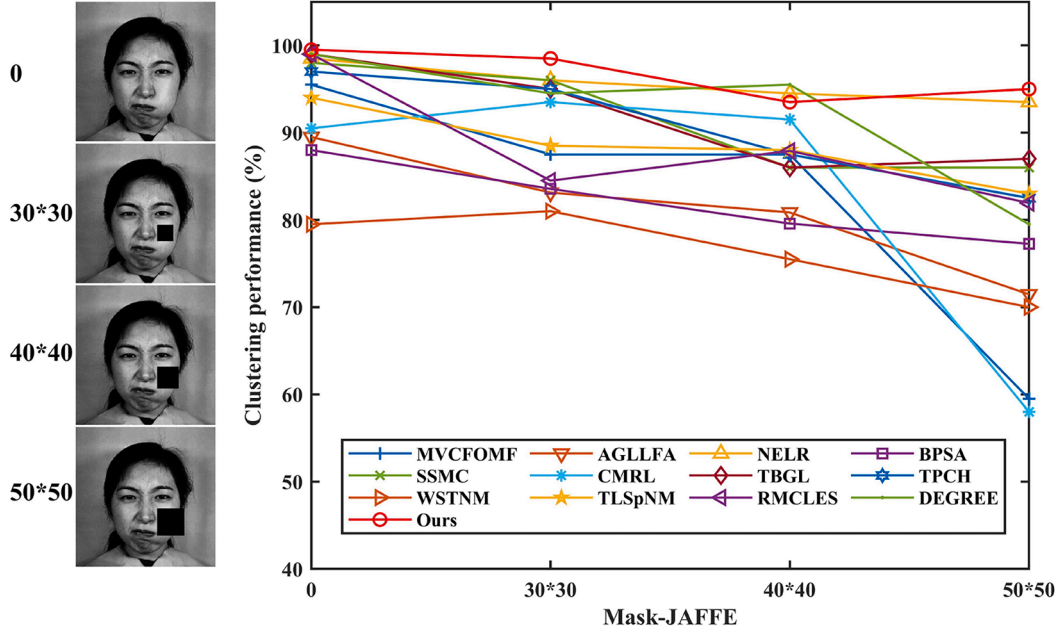


Fig. 10. Clustering performance via different methods on the Mask-JAFFE dataset with different mask magnitudes (30*30, 40*40, and 50*50).

across views. Furthermore, the introduced ESR term incorporates multi-structural information among views, resulting in more comprehensive and effective feature representations.

4. Optimization and analysis

4.1. Optimization

By introducing $\mathcal{Z} = \mathcal{J}$ into (25), we can derive the augmented Lagrange function:

$$\begin{aligned}
 \min \quad & \|\mathcal{J}\|_{NHTR} + \sum_{v=1}^m \left\| I - H^{(v)} \right\|_F^2 + \alpha \|E\|_{2,1} + \beta \|\mathcal{Z}\|_{ESR} \\
 & + \frac{\mu}{2} \sum_{v=1}^m \left\| X^{(v)} H^{(v)} + \frac{A_1^{(v)}}{\mu} \right\|_F^2 \\
 & + \frac{\mu}{2} \sum_{v=1}^m \left\| H^{(v)} - H^{(v)} Z^{(v)} - E^{(v)} + \frac{A_2^{(v)}}{\mu} \right\|_F^2 \\
 & + \frac{\varphi}{2} \left\| \mathcal{Z} - \mathcal{J} + \frac{\mathcal{W}}{\varphi} \right\|_F^2 \\
 \text{s.t.} \quad & Z^{(v)} \geq 0,
 \end{aligned} \tag{26}$$

where $A_1^{(v)}$, $A_2^{(v)}$ and $\mathcal{W}$ are the Lagrange multipliers, μ and φ are the penalty parameters. Then, we can adopt ADMM to alternately optimize the variables.

(1) For the variable $H^{(v)}$, the corresponding optimization subproblem is as follows

$$\begin{aligned}
 \min \quad & \sum_{v=1}^m \left\| I - H^{(v)} \right\|_F^2 + \frac{\mu}{2} \sum_{v=1}^m \left\| X^{(v)} H^{(v)} + \frac{A_1^{(v)}}{\mu} \right\|_F^2 \\
 & + \frac{\mu}{2} \sum_{v=1}^m \left\| H^{(v)} - H^{(v)} Z^{(v)} - E^{(v)} + \frac{A_2^{(v)}}{\mu} \right\|_F^2
 \end{aligned} \tag{27}$$

Then, we have

$$\begin{aligned}
 & \mu(X^{(v)})^T X^{(v)} H^{(v)} + H^{(v)} (\mu M_1^{(v)} (M_1^{(v)})^T + 2I) \\
 & = -(X^{(v)})^T A_1^{(v)} + 2I + \mu H^{(v)} M_2^{(v)} (M_1^{(v)})^T
 \end{aligned} \tag{28}$$

where $M_1^{(v)} = I - Z^{(v)}$, $M_2^{(v)} = E^{(v)} - \frac{A_2^{(v)}}{\mu}$. Note that (28) is a standard Sylvester problem, and it can be solved by using the Sylvester function available in MATLAB.

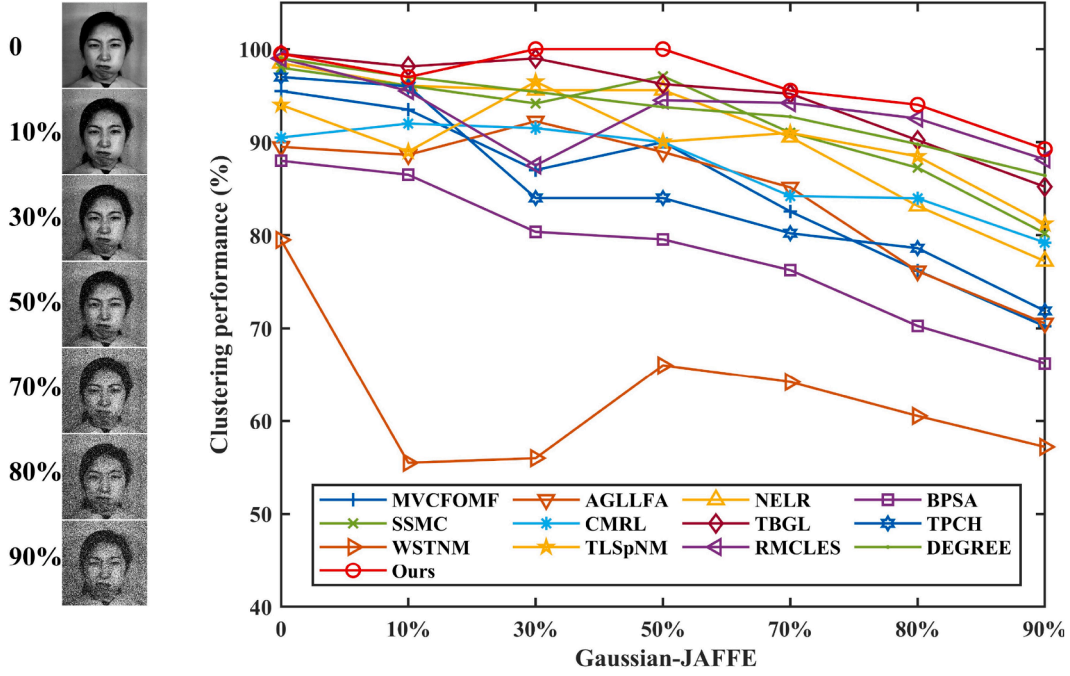


Fig. 11. Clustering performance via different methods on the Gaussian-JAFFE dataset with varying noise ratios (10%, 30%, 50%, 70%, 80%, and 90%).

(2) For the variable $Z^{(v)}$, the corresponding optimization subproblem is as follows

$$\min_{Z^{(v)}} \beta \text{tr}((Z^{(v)})^T((1-v)L^{(v)} + vM)Z^{(v)}) + \frac{\mu}{2} \left\| H^{(v)} - H^{(v)}Z^{(v)} - E^{(v)} + \frac{A_2^{(v)}}{\mu} \right\|_F^2 + \frac{\varphi}{2} \left\| Z^{(v)} - J^{(v)} + \frac{W^{(v)}}{\varphi} \right\|_F^2 \quad (29)$$

s.t. $Z^{(v)} \geq 0$

Taking the partial derivative of $Z^{(v)}$ in (29), we obtain

$$Z^{(v)} = (2\beta M_e + \mu(H^{(v)})^T H^{(v)} + \varphi I)^{-1} (\mu(H^{(v)})^T M_3^{(v)} + \varphi M_4^{(v)}) \quad (30)$$

where $M_e = (1-v)L^{(v)} + vM$, $M_3^{(v)} = H^{(v)} - E^{(v)} + \frac{A_2^{(v)}}{\mu}$, $M_4^{(v)} = J^{(v)} - \frac{W^{(v)}}{\varphi}$.

(3) For the variable E , the corresponding optimization subproblem is as follows

$$\min_E \frac{1}{2} \left\| E - M_5 \right\|_F^2 + \frac{\alpha}{\mu} \|E\|_{2,1} \quad (31)$$

where $M_5 = \sum_{v=1}^m \left\| H^{(v)} - H^{(v)}Z^{(v)} + \frac{A_2^{(v)}}{\mu} \right\|_F^2$. As noted in [34], we can obtain the optimal solution as

$$E_{:,i}^{(v)} = \begin{cases} (1 - \frac{\alpha/\mu}{\|(M_5)_{:,i}\|_2})(M_5)_{:,i}, & (M_5)_{:,i} > \frac{\alpha}{\mu} \\ 0, & \text{otherwise} \end{cases} \quad (32)$$

(4) For the variable J , the corresponding optimization subproblem is as follows

$$\min_J \frac{1}{2} \left\| J - \left(Z + \frac{W}{\varphi} \right) \right\|_F^2 + \frac{1}{\varphi} \|J\|_{NHTR} \quad (33)$$

For this particular optimization problem, the following theory can be used to obtain the solution.

Theorem 3. Given a third-order tensor $\mathcal{K} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, where $\mathcal{K} = \mathcal{U} * S * \mathcal{V}^T$ and $\vartheta > 0$. There exists a Normalized Hyperbolic Tensor Rank problem that can be written as

$$\arg \min_J \frac{1}{2} \left\| J - \mathcal{K} \right\|_F^2 + \vartheta \|J\|_{NHTR} \quad (34)$$

The optimal solution for this problem is

$$J = \mathcal{U} * \text{ifft}(\text{Prox}_{f,\vartheta}(S_f), [], 3) * \mathcal{V}^T. \quad (35)$$

where $\text{ifft}(\text{Prox}_{f,\vartheta}(S_f), [], 3)$ is an f -diagonal tensor, and $\text{Prox}_{f,\vartheta}(S_f^k(i, i)) = \arg \min_{x \geq 0} \frac{1}{2} (x - S_f^k(i, i))^2 + \vartheta f(x)$ with $f(x) = \text{Nortanh}(x)$.

As described in [18], Theorem 3 has been proved. Obviously, $\text{Prox}_{f,\vartheta}(S_f^k(i, i))$ is obtained by combining convex and concave functions. And such functions can be solved by utilizing the differential iteration of convex (DC) (Tao & An, 1997) programming until convergence is achieved, resulting in a closed-form solution.

$$\theta^{\text{iter}+1} = \left(S_f^k(i, i) - \frac{\partial f(\theta^{\text{iter}})}{\varphi} \right)_+ \quad (36)$$

where iter is the number of iterations, $\theta = \text{Prox}_{f,\vartheta}(S_f^k(i, i))$.

(5) For other variables, these are updated as follows

$$\begin{cases} \mathcal{W} = \mathcal{W} + \varphi(\mathcal{Z} - J) \\ A_1^{(v)} = A_1^{(v)} + \mu X^{(v)} H^{(v)} \\ A_2^{(v)} = A_2^{(v)} + \mu(H^{(v)} - H^{(v)}Z^{(v)} - E^{(v)}) \\ \mu = \min(\tau\mu, \mu_{\max}), \varphi = \min(\tau\varphi, \varphi_{\max}) \end{cases} \quad (37)$$

where τ is an automatically adjusted parameter used to control the size between $\mu(\varphi)$ and $\mu_{\max}(\varphi_{\max})$. According to the above derivation, the details of the proposed method are summarized in Algorithm 1.

According to the above derivation, the details of the proposed method are summarized in Algorithm 1.

4.2. Complexity analysis

The main computational complexity of the proposed method is to solve subproblems $H^{(v)}$, $Z^{(v)}$, $E^{(v)}$ and J . Table 3 lists the complexity of each subproblem and total complexity, where iter is the number of iterations.

4.3. Connections to other methods

In this section, we primarily compare the similarities and differences between the proposed method and two related approaches: Learning

Algorithm 1 Dimension-adaptive latent representation learning with normalized hyperbolic tensor rank.

Input: multi-view data $\{X^{(1)}, \dots, X^{(v)}\}$, maximum iterations, all hyperparameters.
1: Initialization: initialize $\mathcal{Z}$, $\mathcal{J}$ and $\mathcal{W}$ to zero tensor, $A_1^{(v)}, A_2^{(v)}$ to zero matrix;
 set $\tau = 2$, $\mu = \varphi = 10^{-5}$, $\mu_{\max} = \varphi_{\max} = 10^{12}$
2: while not converged do
3: update $U^{(v)}$ by (28);
4: update $Z^{(v)}$ by (30);
5: update $E^{(v)}$ by (32);
6: update $\mathcal{J}$ by (36);
7: update others by (37);
8: check convergence: $\|H^{(v)} - H^{(v)}Z^{(v)} - E^{(v)}\|_F^2 \leq 10^{-7}$
9: end while
Output: optimal similarity graph $Z^{(v)}$.

Table 3

Computational complexity of subproblems about the proposed method.

Subproblems	$H^{(v)}$	$Z^{(v)}$	E	$\mathcal{J}$
Complexity	$\mathcal{O}(n^3)$	$\mathcal{O}(n^3)$	$\mathcal{O}(n^2)$	$\mathcal{O}(mn^2 \log(mn) + n^2 m^2)$
Total Complexity	$\mathcal{O}(\text{iter}(mn^3 + mn^2 + mn^2 \log(mn) + n^2 m^2))$			

Latent Disentangled Embeddings and Graphs for Multi-View Clustering (DEGREE) (Zhang et al., 2024) and Partition-Level Fusion Induced Multi-View Subspace Clustering with Tensorial Geman Rank (PFMSC) (Zhao et al., 2024).

Compared with DEGREE: The DEGREE model is as follows

$$\begin{aligned}
 \min \quad & \sum_{\substack{i,j=1 \\ i < j, i \neq j}}^m (\|H^T D^{(v)}\|_F^2 + \|(D^{(v)})^T D^{(v)}\|_F^2) + \beta \|Z\|_* + \gamma \|\mathcal{Q}\|_{\otimes}^2 \\
 & + \sum_{i=1}^m \|E_1^{(v)}\|_{2,1} + \lambda (\|E_2\|_{2,1} + \sum_{i=1}^m \|E_3^{(v)}\|_{2,1}) \\
 \text{s.t.} \quad & \begin{cases} X^{(v)} = P^{(v)}(H + D^{(v)}) + E_1^{(v)}, \\ H = HZ + E_2, \quad D^{(v)} = D^{(v)}Q^{(v)} + E_3^{(v)}, \\ (P^{(v)})^T P^{(v)} = I, \quad \mathcal{Q} = \Phi(Q^{(1)}, Q^{(2)}, \dots, Q^{(m)}). \end{cases}
 \end{aligned} \quad (38)$$

where $D^{(v)} \in \mathbb{R}^{d_i \times n}$ means the unique information of each view and $H \in \mathbb{R}^{d_i \times n}$ is the latent consistent representation of all views, $P^{(v)} \in \mathbb{R}^{d_v \times d_l}$ represents the projection matrix of each view and d_l is the predefined dimension of the low-dimensional subspace, $Z \in \mathbb{R}^{n \times n}$ and $Q^{(v)} \in \mathbb{R}^{n \times n}$ are similarity matrices representing consistency and diversity respectively.

Both DEGREE and DLRL-NHTR are multi-view clustering methods based on latent representation learning, in which the latent representation is exploited as a self-representation dictionary for clustering. The major difference between them is that DLRL-NHTR does not require the dimensionality to be predefined when learning the latent representation, thus exhibiting stronger adaptability. As shown in Section 3.2, the size of the representation space in our method is automatically determined during the learning process. By contrast, DEGREE depends on a manually predefined latent dimensionality, which limits its flexibility and generalization capability to a certain extent. Furthermore, DEGREE is less effective in characterizing the intra-view structural information, especially in capturing the global and local structures within each view.

Compared with PFMSCL: The PFMSCL model is as follows

$$\begin{aligned}
 \min \quad & \|\mathcal{Z}\|_{TGR} + \lambda \|E\|_{2,1} + \alpha \sum_{v=1}^m \frac{1}{w^{(v)}} \text{tr}(F^T L_{U^{(v)}} F) \\
 \text{s.t.} \quad & \begin{cases} X^{(v)} = X^{(v)}Z^{(v)} + E^{(v)}, \quad v = 1, 2, \dots, m, \\ \mathcal{Z} = \Phi(Z^{(1)}, Z^{(2)}, \dots, Z^{(m)}), \quad E = [E^{(1)}; E^{(2)}; \dots; E^{(m)}], \\ w^T \mathbf{1} = 1, \quad F^T F = 1. \end{cases}
 \end{aligned} \quad (39)$$

where $U^{(v)} = \frac{|Z^{(v)}| + |(Z^{(v)})^T|}{2}$ means the unified affinity matrix, $L_{U^{(v)}} = D^{(v)} - Z^{(v)}$ the Laplace matrix with $D_{i,i}^{(v)} = \sum_{j=1}^n Z_{i,j}^{(v)}$, $F \in \mathbb{R}^{n \times c}$ is the indicator matrix and c is the number of clusters, $w^{(v)}$ is a weight value of v th view.

Both PFMSCL and DLRL-NHTR distinguish the importance of different singular values by constructing novel rank approximation functions, thereby more thoroughly revealing the underlying high-order structure of multi-view data. Compared with PFMSCL, DLRL-NHTR exhibits advantages in the following three aspects. First, for intra-view structural information analysis, while PFMSCL mainly focuses on local information, our method performs a collaborative analysis from both global and local perspectives, thereby enhancing the representation capability for intra-view structural characteristics. Second, our method employs latent representations with adaptive dimensionality as the dictionary for self-representation learning, without requiring the subspace dimensionality to be predefined, thus improving the adaptability of the model. Third, according to the comparative analysis of the properties of NHTR and TGR, NHTR can approximate the true rank more accurately in the region of relatively large singular values, which is more conducive to characterizing the low-rank properties of data.

5. Experiments

In this section, we selected ten benchmark datasets to test our proposed method and compared it with several related methods. These datasets are divided into multiple categories to ensure diversity in the experiment environment, thereby fully validating the feasibility of the method.

5.1. Experimental setting

Datasets: In the experiment, we used ten different benchmark datasets to perform testing, including NGS, COIL20, BBC, BBCSports (BBCs), JAFFE, CVLAB, PIE, CMUPIE, UCI, and SCENE15. These datasets are summarized in Table 4.

Compared methods: To demonstrate the superiority of the proposed method, we compare it with several state-of-the-art clustering methods, including:

(1) Subspace-based multi-view clustering (SMVC) methods: Multi-view clustering using a flexible and optimal multi-graph fusion method (MVCFOF) (Kan et al., 2024), Multi-view Subspace Clustering via Adaptive Graph Learning and Late Fusion Alignment (AGLLFA) (Tang et al., 2023), Robust multi-view subspace clustering via neighbor embedding on manifold and low-rank representation learning (NELR) (Kong et al., 2025), and Bidirectional probabilistic subspaces approximation for multiview clustering (BPSA) (Wu et al., 2022).

(2) Latent representation multi-view clustering (LRMVC) methods: Relaxed multi-view clustering in latent embedding space (RMCLES) (Chen et al., 2021), Joint learning of latent subspace and structured graph for multi-view Clustering (SSMC) (Wang et al., 2024), Comprehensive Multi-view Representation Learning (CMRL) (Zheng et al., 2023), and Learning latent disentangled embeddings and graphs for multi-view clustering (DEGREE) (Zhang et al., 2024).

(3) Tensor-based multi-view clustering (TMVC) methods: Multiview Subspace Clustering by an Enhanced Tensor Nuclear Norm (WSTNM)

Table 4
The details of the datasets.

Dataset	Type	Dimension	Views/Classes/Instances
NGS	Text	1323/867/3072/1083	3/5/500
COIL20	Object	1024/944/4096/576	4/20/1440
BBC	Text	4659/4633/4565/4684	4/5/685
BBCs	Text	3183/3203	2/5/544
JAFFE	Image	1369/1089/4096/1225	4/10/200
CVLAB	Image	1323/867/3072/1083	4/20/600
CMUPIE	Image	1024/3102/2060	3/68/2856
PIE	Image	484/256/279	3/68/1020
UCI	Digit	240/76/6	3/10/2000
SCENE15	Scene	1800/1180/1240	3/15/4485

(Xia et al., 2021), Tensor-interacted Projection and Cooperative Hashing for Multi-view Clustering (TPCH) (Wang et al., 2025), Logarithmic Schatten-p norm minimization for tensorial multi-view subspace clustering (TLSpNM) (Guo et al., 2022), and Tensorized bipartite graph learning for multi-view clustering (TBGL) (Xia et al., 2022).

Evaluation metrics: In the experiment, we adopted seven different evaluation metrics. These metrics include Accuracy (ACC), Normalized Mutual Information (NMI), Purity (PUR), Fscore, Precision, Recall, and Adjusted Rand Index (ARI). These metrics range from 0 to 1, where values closer to 1 indicate better clustering performance.

5.2. Experimental results and analysis

The experimental results of all methods are summarized in Tables 5 to 6 with the best results and second-best results indicated by bold values and underlines, respectively. From these experimental results, we can draw the following conclusions.

The experiment results indicate that the proposed method achieved the best performance across seven clustering metrics on most of the datasets, with a few metrics ranking second best. Particularly, on the NGS dataset, the method obtained perfect scores across all metrics. Moreover, the proposed method demonstrated significant performance improvements on some datasets. For instance, on the BBCs dataset, it surpassed the second-best method by 2.94%, 7.22%, 2.94%, 4.97%, 3.03%, 3.48%, and 8.54% across the seven metrics, respectively. Although the clustering scores on the CVLAB dataset were mostly second-best, the performance gap relative to the best CMRL method was minimal, with the maximum difference across the seven metrics not exceeding 0.68%. This small difference does not reduce the superiority and effectiveness of the proposed method. Moreover, on the large-scale SCENE15 dataset, our method achieves the best performance on six clustering metrics except for PUR and attains the second-best result on the PUR metric, with a difference of only 0.23% from the optimal method.

Compared with subspace methods, LRMVC and TMVC methods generally achieve superior clustering performance. Their advantages are primarily reflected in two aspects. First, LRMVC methods introduce latent representations to effectively suppress noise interference and better capture the underlying structural features of the data. Second, TMVC methods exploit the tensor representation to characterize the high-order correlations among multiple views, thereby providing a more comprehensive understanding of the intrinsic distribution patterns of the data. By exploring the essential structure from a broader perspective, these methods often produce higher-quality clustering results. However, both LRMVC and TMVC methods also suffer from some limitations. For LRMVC methods, determining an appropriate dimensionality for the latent representation is an important factor that limits their applicability to more complex and diverse scenarios. For TMVC methods, problems such as unbalanced singular value penalties or insufficient penalty strength may lead to low-quality tensor representation, thus reducing clustering performance. These limitations also help explain why LRMVC and TMVC methods may underperform compared with some subspace methods on some datasets.

In contrast, our method addresses the limitations mentioned above. First, we address the limitation of traditional latent representation learning, which is highly dependent on dimension selection. By learning latent representations in a specific null space, our method not only captures intrinsic data correlations more accurately but also avoids the uncertainty introduced by manual parameter tuning. Second, we design a novel tensor rank approximation strategy that adjusts the penalty strength according to the importance of different singular values, leading to more precise tensor rank approximation. Finally, we comprehensively integrate multi-structural information, which significantly enhances the discriminative ability and robustness of the representation matrices. Thus, these factors contribute to the significant advantages of our method in clustering performance.

To further verify the superiority of the proposed method, using ACC as an example, we performed the Bayesian Signed Rank test on all datasets. The results are shown in Fig. 3, and each square shows the probability that the method in the row is better than the method in the column. A value closer to 1 means a stronger advantage. As shown, the probability that LRMVC and TMVC perform better than MVC is higher than 50% in most cases, which clearly shows the superiority of these methods. It is also observed that on JAFFE and CVLAB, the probability of the WSTNM method being better than other methods is close to 0. This may be because it is difficult to accurately approximate the tensor rank of the data, which results in lower performance. Overall, the proposed method shows a high probability close to 1 on most datasets, which shows that it can effectively learn latent representations that contain complete information and can more accurately capture the higher-order correlations of the data.

5.3. Model analysis

Parameter analysis: In our model, it involves three parameters α , β , and γ . For α and β , we first empirically set both search ranges as $\{10^{-4}, 10^{-3}, \dots, 10^4\}$. Fig. 4 shows the ACC corresponding to different parameter settings while γ is fixed. The results indicate that our method achieves robust and excellent experimental performance within relative parameters. For unsupervised learning, determining the optimal parameters remains an issue that needs to be addressed, and it is not necessary to determine the specific values of each parameter. Therefore, we narrowed down the range of these two parameters based on the results in Fig. 4, i.e., $\alpha \in \{10^{-5}, 10^{-4}, \dots, 1\}$ and $\beta \in \{10^{-5}, 10^{-4}, \dots, 10\}$.

Then, we fixed the parameters α and β to verify the influence of the rank parameter γ . Fig. 5 shows the ACC for different values of γ . From Fig. 5, we observe that the clustering performance is sensitive to changes in γ , because γ directly controls how much different singular values are penalized. As for γ , the specific values used in the experiment are listed in Table 7.

Normalized Hyperbolic Tensor Rank analysis: The experimental results in Section 6.2 have already demonstrated the superiority of our method under different MVSC frameworks. In this section, we replace NHTR in our method with TNN, TLSpN, and TGR to compare the performance of different tensor rank approaches within the same framework. Fig. 6 presents the experimental results across ten datasets. The results show that the DLRL-TNN method achieves the poorest clustering performance on all datasets, indicating the necessity of distinguishing different singular values to assess their importance. Although the DLRL-TLSpN method improves upon DLRL-TNN, its inability to properly constrain large singular values still limits its clustering performance. In contrast, both DLRL-NHTR and DLRL-TGR can effectively analyze differences among singular values and apply differentiated penalties accordingly, resulting in comparable performance on most datasets. However, on the PIE, UCI, and SCENE15 datasets, the performance of DLRL-TGR is slightly lower than that of DLRL-NHTR. As analyzed in Section 3.2, TGR strategy exhibits lower approximation accuracy to the true rank for large singular values, which explains the superior performance of DLRL-NHTR over DLRL-TGR. Overall, NHTR strategy is able to apply more appropriate

Table 5
Experiment results on NGS, JAFFE, CVLAB, BBC, and BBCs.

Dataset	Metric	MVCFOMF	AGLLFA	NELR	BPSA	SSMC	CMRL	RMCLES	TBGL	DEGREE	WSTNM	TLSpN	TPCH	Ours
NGS	ACC	72.60	44.49	89.40	63.20	98.00	96.00	96.40	97.80	97.20	74.20	<u>98.20</u>	90.60	100
	NMI	49.45	33.74	74.49	55.37	93.22	89.87	88.80	92.80	91.17	45.13	<u>94.60</u>	80.44	100
	PUR	72.60	45.02	89.40	67.80	98.00	96.00	96.40	97.80	87.20	74.20	<u>98.20</u>	90.60	100
	Fscore	58.84	43.50	80.72	55.62	96.03	92.29	92.91	95.64	94.44	56.56	<u>96.45</u>	82.71	100
	Precision	57.39	32.03	80.68	44.50	96.01	92.18	92.87	<u>96.63</u>	94.38	56.07	<u>96.43</u>	81.92	100
	Recall	60.37	70.73	80.76	74.16	96.04	92.40	92.95	<u>95.65</u>	84.50	57.05	<u>96.48</u>	83.52	100
	ARI	48.33	22.18	75.95	40.99	95.04	90.38	91.16	94.56	93.06	45.69	<u>95.57</u>	78.38	100
JAFFE	ACC	95.50	89.50	98.50	88.00	98.00	90.50	99.00	<u>99.48</u>	99.00	79.50	94.50	97.00	99.50
	NMI	94.40	91.69	97.69	92.96	97.19	60.62	98.55	<u>99.07</u>	98.25	78.69	97.28	95.85	99.13
	PUR	85.50	90.67	98.50	89.50	98.00	90.50	99.00	<u>99.47</u>	99.00	79.50	97.50	97.00	99.50
	Fscore	92.07	86.22	97.03	87.77	96.06	84.97	98.00	<u>98.90</u>	97.95	69.42	95.43	94.22	98.97
	Precision	91.88	83.84	96.75	80.62	98.81	84.42	97.90	<u>98.93</u>	97.90	68.10	94.80	94.03	98.95
	Recall	92.26	88.92	97.10	96.32	96.32	85.53	98.10	<u>98.97</u>	98.00	70.79	96.05	94.42	99.00
	ARI	91.23	84.70	96.71	86.35	95.65	83.37	97.79	99.87	97.73	66.12	94.94	93.61	98.87
CVLAB	ACC	81.92	85.19	92.67	93.00	92.50	99.33	77.83	93.00	88.83	71.17	96.83	93.83	<u>99.00</u>
	NMI	94.44	94.84	94.88	97.69	97.69	99.31	89.03	97.69	92.75	76.06	96.67	98.38	<u>99.10</u>
	PUR	84.70	88.96	92.67	95.00	95.00	99.33	81.00	95.00	88.83	74.50	96.83	95.00	<u>99.00</u>
	Fscore	79.31	85.60	89.00	93.83	93.78	98.71	76.67	93.83	84.41	60.43	93.96	94.15	<u>98.14</u>
	Precision	69.07	80.42	88.28	90.41	90.40	98.62	71.02	90.41	82.98	58.57	93.57	90.46	<u>97.94</u>
	Recall	96.70	91.70	89.72	97.52	97.41	98.80	83.30	97.52	85.90	62.41	94.36	98.15	<u>98.34</u>
	ARI	77.96	84.81	88.43	93.50	93.75	98.65	75.39	93.50	83.60	58.35	93.65	93.84	<u>98.05</u>
BBC	ACC	86.92	52.47	91.53	71.97	89.05	93.58	86.42	66.28	91.68	88.18	<u>93.72</u>	84.53	96.64
	NMI	66.36	31.44	78.07	53.18	74.46	86.64	68.83	45.43	77.35	79.42	<u>86.63</u>	69.83	88.37
	PUR	86.92	54.77	91.53	72.26	89.05	93.58	86.42	67.01	91.68	88.18	<u>93.72</u>	84.53	96.64
	Fscore	78.53	44.25	86.08	67.15	81.44	88.78	79.99	61.65	85.70	81.27	<u>89.12</u>	77.94	93.61
	Precision	78.52	35.90	84.02	54.71	76.50	92.84	77.66	48.49	85.52	86.82	<u>93.27</u>	82.02	94.09
	Recall	78.55	57.71	<u>88.24</u>	86.91	87.07	85.06	82.46	84.64	85.89	76.40	<u>85.33</u>	74.24	93.14
	ARI	71.95	21.62	<u>81.68</u>	53.86	75.27	85.54	73.61	45.35	81.31	76.00	<u>85.98</u>	71.61	91.67
BBCs	ACC	94.00	46.83	92.46	69.30	71.14	<u>95.96</u>	93.20	54.04	81.80	75.18	94.67	82.17	98.90
	NMI	84.19	26.03	80.52	61.03	67.60	<u>89.08</u>	81.30	25.53	67.04	66.48	88.00	84.05	96.30
	PUR	94.02	51.51	92.46	73.35	75.00	<u>95.96</u>	93.20	54.04	81.80	82.35	94.67	86.77	98.90
	Fscore	89.52	41.36	85.11	66.91	71.10	<u>92.67</u>	86.62	46.70	71.23	74.91	90.67	85.62	97.64
	Precision	91.62	32.74	86.36	52.51	57.38	<u>94.84</u>	86.62	31.48	75.02	75.04	93.00	87.25	97.87
	Recall	87.57	56.17	83.88	92.17	<u>93.42</u>	90.60	86.63	90.40	67.80	74.78	88.46	84.06	96.90
	ARI	86.31	16.07	80.53	52.46	58.97	<u>90.44</u>	82.44	17.52	62.80	67.07	87.84	81.23	96.38

penalties to different singular values, thereby producing a tensor representation with stronger discriminative capability.

Ablation analysis: In this section, we use ablation experiments to verify the validity and effectiveness of the proposed method. As shown in Table 8, three simplified versions of the model are constructed by removing different regularization terms: **Model 1** does not use the NHTR term; **Model 2** ignores the structural information in the original view; **Model 3** removes the latent representation learning form. Fig. 7 shows the clustering results of the three degraded models and the complete model. It can be observed that the clustering performance of all simplified models is much lower than that of the complete model, which shows that each module plays an important role in improving the clustering performance.

The complexity analysis: In Tables 9 and 10, we report the running time across different datasets and theoretical computational complexity of each method, where k denotes the number of clusters, $iter$ means iteration numbers, and d_l represents the predefined dimensionality of the low-dimensional subspace. In addition, bold, underlined, and italic fonts respectively denote the best, second-best, and worst running times. Overall, the differences in running time across datasets are influenced not only by theoretical complexity but also by practical factors such as the number of convergence iterations and the sparsity of matrices. First, TPCH combines bipartite graph and anchor graph strategies, reducing the computational complexity to $\mathcal{O}(\log n)$. This allows TPCH to maintain consistently high computational efficiency and stability across datasets of varying sizes, with running time remaining low even as the number of samples increases. It is noteworthy that, on smaller datasets, although MVCFOMF and AGLLFA have theoretical complexities of $\mathcal{O}(n^3)$, the actual running time of TPCH is slightly higher than that of AGLLFA

(the fastest) and MVCFOMF (the second fastest). This phenomenon is primarily due to the fact that MVCFOMF and AGLLFA impose sparsity constraints on the representation coefficient matrices, and most computations are performed on low-dimensional or sparse matrices without large-scale singular value or tensor decompositions, resulting in very short running times when the number of samples is relatively small. However, on larger datasets such as UCI, CMUPE, and SCENE15, TPCH significantly outperforms the other two methods. Furthermore, for low-rank tensor-based latent representation multi-view subspace clustering methods, the theoretical complexity is also at the $\mathcal{O}(n^3)$ level, but singular value decomposition and tensor decomposition operations increase the actual running time. Overall, our method achieves a moderate running time across datasets of different scales, striking a reasonable balance between clustering accuracy and computational efficiency.

Convergence analysis: In this section, we verify the convergence of our method through experiments. We use $\|H^{(v)} - H^{(v)}Z^{(v)} - E^{(v)}\|_F^2 \leq 10^{-7}$ as the convergence termination condition, and the convergence results on nine datasets are shown in Fig. 8. From the Fig. 8, we can see that our method achieves convergence on all datasets, and the convergence number is less than the pre-set 50 iterations, indicating a fast convergence speed.

Robustness analysis: In this section, we constructed two noisy datasets based on JAFFE to further evaluate the robustness of the proposed method. Specifically, Gaussian noise with varying ratios (10%, 30%, 50%, 70%, 80%, and 90%) and mask noise with different mask magnitudes (30*30, 40*40, and 50*50) were introduced into the raw JAFFE dataset. Representative examples of these noisy datasets are shown in Fig. 9. The clustering results of all compared methods on the two noisy datasets are reported in Figs. 10 and 11. As illustrated in Figs. 10 and 11,

Table 6
Experimental results on CMUPIE, COIL20, PIE, UCI, and SCENE15.

Dataset	Metric	MVCFOMF	AGLLFA	NELR	BPSA	SSMC	CMRL	RMCL	TBGL	DEGREE	WSTNM	TLSpNM	TPCH	Ours
CMUPIE	ACC	82.45	72.40	61.80	77.31	72.37	81.02	32.46	60.43	90.48	85.12	87.71	46.81	91.07
	NMI	95.49	89.14	82.07	90.39	88.80	91.58	61.04	78.03	97.07	91.50	96.23	69.03	96.99
	PUR	86.92	75.67	65.13	81.72	77.00	82.28	33.37	67.68	<u>93.21</u>	87.01	90.76	47.83	93.49
	Fscore	84.05	69.51	55.34	70.25	66.63	77.62	19.35	34.41	90.64	79.15	88.13	34.49	90.38
	Precision	75.91	64.87	51.89	62.29	60.46	75.24	12.28	22.91	87.54	75.88	83.29	31.33	<u>87.24</u>
	Recall	84.27	74.91	59.29	80.54	74.21	80.15	45.51	69.07	93.87	82.72	93.56	38.36	93.75
	ARI	83.79	69.04	54.65	69.76	66.09	77.28	17.48	32.97	90.50	78.84	87.95	33.44	<u>90.23</u>
COIL20	ACC	61.05	63.82	62.50	76.46	78.40	83.54	75.07	72.50	80.90	65.83	97.50	60.21	<u>93.40</u>
	NMI	72.44	76.81	79.87	87.72	87.61	89.76	82.06	85.91	88.24	70.31	<u>93.03</u>	72.06	96.69
	PUR	64.19	68.02	70.00	82.92	82.64	83.68	75.21	78.44	83.47	70.28	<u>89.44</u>	64.86	94.38
	Fscore	53.64	60.19	56.17	67.27	73.50	82.57	70.07	62.34	78.92	57.96	<u>86.17</u>	54.51	93.18
	Precision	49.23	56.25	43.91	53.91	65.74	82.16	67.82	47.22	77.26	56.59	<u>84.66</u>	52.12	89.50
	Recall	59.00	64.85	77.92	<u>89.43</u>	83.33	82.97	72.47	91.69	80.66	59.41	87.72	57.12	87.18
	ARI	51.00	<u>87.96</u>	53.22	65.12	71.95	81.66	68.46	59.72	77.81	55.73	85.43	52.03	92.82
PIE	ACC	84.65	25.38	22.55	42.84	38.43	50.39	42.25	27.45	85.69	75.98	<u>89.90</u>	47.26	90.10
	NMI	89.60	55.13	53.42	67.14	65.62	74.13	63.54	38.74	90.80	85.75	<u>95.62</u>	69.85	96.50
	PUR	85.42	26.70	25.39	46.27	42.35	52.84	44.61	33.14	86.86	77.45	<u>91.08</u>	49.22	92.35
	Fscore	73.65	13.83	11.37	23.93	24.35	39.76	24.68	4.49	70.15	65.66	<u>87.61</u>	29.91	89.08
	Precision	69.56	11.90	10.07	17.62	21.90	38.14	19.06	2.38	60.73	63.04	<u>84.60</u>	27.26	85.40
	Recall	78.28	16.52	13.04	37.13	27.42	41.53	34.96	40.46	83.01	68.52	<u>90.83</u>	33.12	93.08
	ARI	73.26	12.43	9.97	22.48	23.18	38.89	23.31	1.94	69.66	65.17	<u>87.43</u>	28.83	88.92
UCI	ACC	73.15	82.29	53.40	95.30	86.95	<u>97.35</u>	66.45	84.80	94.85	56.45	95.70	73.85	97.60
	NMI	69.10	74.75	47.39	90.42	87.14	<u>93.97</u>	66.77	87.74	89.82	39.11	89.54	84.11	94.34
	PUR	75.50	83.29	56.50	95.30	86.95	<u>97.35</u>	66.50	87.25	94.85	58.25	98.70	78.95	97.60
	Fscore	65.00	72.61	42.58	90.92	83.77	<u>94.82</u>	58.55	83.73	90.15	40.69	91.60	70.95	95.27
	Precision	63.20	71.42	39.87	90.84	83.23	<u>94.80</u>	54.18	77.72	89.96	40.20	92.58	60.86	95.24
	Recall	66.91	73.86	45.70	91.00	84.33	<u>94.83</u>	63.68	90.75	90.33	41.19	91.62	85.06	95.30
	ARI	61.01	69.52	35.75	89.92	81.97	<u>94.24</u>	53.55	81.78	89.05	34.04	90.67	67.14	94.75
SCENE15	ACC	36.52	28.78	38.71	39.38	38.28	<u>81.58</u>	80.27	77.54	79.21	36.12	80.47	78.37	82.07
	NMI	34.67	27.33	33.42	37.47	37.13	<u>85.79</u>	79.94	77.08	78.14	31.26	79.77	84.61	88.16
	PUR	39.36	32.78	39.87	42.81	44.41	86.32	79.42	76.98	78.69	40.11	81.32	83.30	86.09
	Fscore	26.51	21.69	26.94	27.25	26.59	83.07	80.75	75.47	78.25	24.84	<u>83.45</u>	76.42	84.60
	Precision	25.40	19.48	25.10	25.36	24.05	84.56	75.63	75.97	77.36	23.20	<u>84.58</u>	74.36	85.54
	Recall	25.99	20.01	25.10	29.45	30.68	84.65	76.78	76.39	76.84	23.69	<u>86.39</u>	78.60	86.79
	ARI	20.10	13.69	19.54	21.42	20.83	<u>84.27</u>	77.64	80.11	77.60	17.70	83.69	74.61	85.37

Table 7
The value of γ on different datasets.

Datasets	NGS	JAFFE	CVLAB	BBC	BBCs	CMUPIE	COIL20	PIE	UCI	SCENE15
Values	1	10	0.1	0.1	1	1	1	10 ²	10 ⁻⁴	1

Table 8
Specific functions of the simplified version.

Models	Corresponding Function
Model 1	$\min \sum_{i=1}^m \ I - H^{(i)}\ _F^2 + \alpha \sum_{i=1}^m \ E^{(i)}\ _{2,1} + \beta \ \mathcal{Z}\ _{ESR}$ $\text{s.t. } X^{(v)} H^{(v)} = 0, H^{(v)} = H^{(v)} Z^{(v)} + E^{(v)}, Z^{(v)} \geq 0, \mathcal{Z} = \Phi(Z^{(1)}, Z^{(2)}, \dots, Z^{(m)}), v \in \{1, 2, \dots, m\}$
Model 2	$\min \ \mathcal{Z}\ _{NHTF} + \sum_{i=1}^m \ I - H^{(i)}\ _F^2 + \alpha \sum_{i=1}^m \ E^{(i)}\ _{2,1}$ $\text{s.t. } X^{(v)} H^{(v)} = 0, H^{(v)} = H^{(v)} Z^{(v)} + E^{(v)}, Z^{(v)} \geq 0, \mathcal{Z} = \Phi(Z^{(1)}, Z^{(2)}, \dots, Z^{(m)}), v \in \{1, 2, \dots, m\}$
Model 3	$\min \ \mathcal{Z}\ _{NHTF} + \alpha \sum_{i=1}^m \ E^{(i)}\ _{2,1} + \beta \ \mathcal{Z}\ _{ESR}$ $\text{s.t. } X^{(v)} = X^{(v)} Z^{(v)} + E^{(v)}, Z^{(v)} \geq 0, \mathcal{Z} = \Phi(Z^{(1)}, Z^{(2)}, \dots, Z^{(m)}), v \in \{1, 2, \dots, m\}$

Table 9
Running time of all methods on the different datasets.

Methods	NGS	JAFFE	CVLAB	BBC	BBCs	COIL20	PIE	UCI	CMUPIE	SCENE15
MVCFOMF	<u>1.33</u>	<u>0.23</u>	<u>1.93</u>	2.89	<u>1.94</u>	<u>18.60</u>	5.08	25.17	114.83	371.47
AGLLFA	0.35	0.12	0.45	0.83	0.29	1.43	1.89	1.80	<u>9.80</u>	53.40
NELR	65.78	399.34	493.55	315.36	387.79	1146.30	1524.73	2812.71	7619.62	13154.3
BPSA	199.11	175.21	255.57	2570.78	216.62	451.30	17.88	17.83	486.13	1605.83
SSMC	252.30	0.65	96.28	18.17	21.14	603.95	272.35	1242.65	3321.12	6862.27
CMRL	15.52	7.43	24.65	50.36	28.56	275.56	56.89	340.76	610.71	3903.50
TBGL	19.05	7.12	25.75	28.34	15.91	89.08	51.55	239.06	603.07	4441.85
TPCH	2.67	0.77	2.30	<u>1.07</u>	2.16	21.25	5.74	<u>12.90</u>	8.88	6.12
WSTNM	16.93	3.94	20.56	43.91	12.88	512.31	27.81	142.87	142.56	984.69
TLSpNM	18.09	3.48	14.13	31.81	15.36	148.60	13.97	115.36	204.37	856.30
RMCL	166.92	8.57	29.08	35.32	19.08	404.80	122.11	1322.86	3829.69	7834.65
DEGREE	70.90	23.52	86.33	168.81	61.27	162.19	118.47	498.70	1324.84	3144.78
Ours	38.67	7.96	18.21	80.21	28.97	496.79	91.13	708.92	1959.68	3758.41

Table 10
Theoretical computational complexity of each method.

Methods	Theoretical computational complexity
MVCFOMF	$\mathcal{O}(\text{iter}(md^{(v)}n^2 + mn^3 + n^2 \log n + k^3))$
AGLLFA	$\mathcal{O}(\text{iter}(mn^3))$
NELR	$\mathcal{O}(\text{iter}(2md^{(v)}n^2 + 2mn^3))$
BPSA	$\mathcal{O}(\text{iter}(m(d^{(v)})^3 + mn(d^{(v)})^3 + mnkd^{(v)} + km^2n^2))$
SSMC	$\mathcal{O}(\text{iter}(mkn^3 + n^2 + md^{(v)}d_i))$
CMRL	$\mathcal{O}(\text{iter}(m3n^3 + mn^2 + n^2 \log n))$
TBGL	$\mathcal{O}(\text{iter}(3mn^3 + mn^2 + n^2 \log n))$
TPCH	$\mathcal{O}(\text{iter}(\log n))$
WSTNM	$\mathcal{O}(\text{iter}(n^3 + mn^2d^{(v)} + mn^2 \log(mn)))$
TLSpNM	$\mathcal{O}(\text{iter}(mn^2 \log(n) + m^2n^2 + md^{(v)}n))$
RMCLCS	$\mathcal{O}(\text{iter}(m(d^{(v)})^2d_i + mn^3 + md_i^3 + mkn^2))$
DEGREE	$\mathcal{O}(\text{iter}(mn^3 + n^2 \log(n) + d_i k^2 + k^3))$
Ours	$\mathcal{O}(\text{iter}(mn^3 + mn^2 + mn^2 \log(mn) + n^2m^2))$

the proposed method consistently achieves superior clustering performance under diverse noise conditions. Moreover, when the noise level is below 50%, our method can generally maintain relatively stable clustering performance. As the noise ratio gradually increases, all methods inevitably exhibit performance degradation to different degrees. This is mainly because high-intensity noise destroys the intrinsic structure of the original data and weakens the similarity relationships among samples, thereby interfering with the clustering results. In contrast, the proposed method still shows a relatively mild performance decline under high-noise conditions, indicating that it can more effectively suppress noise interference and sufficiently exploit the stable structural information embedded in multi-view data. Overall, the proposed method demonstrates superior clustering performance and strong robustness under different noise conditions.

6. Conclusion

This paper proposes a dimension-adaptive latent representation learning with normalized hyperbolic tensor rank for multi-view clustering. Unlike traditional methods, the proposed method does not require predefining the dimension of the latent representation, thereby avoiding interference from manual parameter selection while providing theoretical guarantees. In addition, we introduce a novel tensor rank approximation norm—Normalized Hyperbolic Tensor Rank (NHTR). Compared with the tensor nuclear norms, NHTR imposes more precise constraints on different singular values, thus fully exploiting the correlations and complementarities among multiple views. Moreover, we design an elastic structure regularization term that not only facilitates capturing high-order correlations across views but also emphasizes multi-level structural information within individual views, allowing for a comprehensive analysis of intrinsic relationships in the data. Extensive experiments conducted on nine benchmark datasets demonstrate that our method significantly outperforms various state-of-the-art methods in clustering performance. Nevertheless, similar to other tensor-based methods, the proposed approach still faces the challenge of high computational complexity. In future work, we will try to incorporate anchor graph or bipartite graph strategies to further improve computational efficiency.

CRedit authorship contribution statement

Gang Zhu: Writing – original draft, Software, Methodology, Conceptualization; **Lixin Han:** Writing – review & editing, Validation, Supervision; **Hong Yan:** Writing – review & editing, Resources; **Jun Zhu:** Resources.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Proofs of the least-squares optimal approximation

Proof. For any $y \in \mathbb{R}^n$, considering a least-squares problem

$$\min \|y - N^{(v)}a\|_F^2 \quad (\text{A.1})$$

Let $c \in \text{Null}(X^{(v)}) = \text{span}(N^{(v)})$, so any feasible c can be written as

$$c = N^{(v)}a, \quad a \in \mathbb{R}^{r^*} \quad (\text{A.2})$$

Thus, the problem (A.2) is equivalently transformed into an unconstrained minimization with respect to a .

$$\min \|y - N^{(v)}a\|_F^2 \quad (\text{A.3})$$

Expanding this problem yields

$$\min \|y - N^{(v)}a\|_F^2 = (y - N^{(v)}a)^T (y - N^{(v)}a) = y^T y - 2a^T N^T y + a^T N^T N a \quad (\text{A.4})$$

Taking the derivative with respect to a and setting it to zero, we can obtain $N^T N a = N^T y$. Since $(N^{(v)})^T N^{(v)} = I$, we can further provide $a = (N^{(v)})^T y$. So $c = N^{(v)}a = N^{(v)}(N^{(v)})^T y = H^{(v)}y$. $\square$

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